Exploiting Separability in Multi-Scale Grey-Box Bayesian Optimization

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Abstract

We consider grey-box optimization problems where the decision variables naturally partition into black-box variables x^{BB} (as arguments to an expensive black-box function) and white-box variables x^{WB}, governed by a set of explicit, closed-form equations that also depend on the output of the black-box function. We exploit this separability through a bilevel reformulation: an outer Bayesian optimization (BO) to optimize the scalar objective as a function of x^{BB} alone, while an inner problem solves the white-box subproblem via global optimization. The Gaussian process surrogate used in BO is therefore defined $\mathbb{R}^{n_{\mathrm{BB}}}$ rather than $\mathbb{R}^{n_{\mathrm{WB}}+n_{\mathrm{BB}}}$, and white-box constraints are satisfied exactly whenever the inner optimizer converges to a feasible point—without penalty functions, chance constraints, or moment approximations. On a suite of 13 benchmark problems bilevel BO achieves 14×–277,000× lower regret than full-space BO across 7,410 independent optimization runs, with comparable wall clock time for 12 of 13 problems. The advantage grows with dimension—up to 277,000× on 5D problems—and is robust to initialization set size and exploration parameters.

Keywords: Bayesian optimization, Grey-box optimization, Bilevel optimization, Surrogate-based optimization, Multi-scale design, Dimensionality reduction

1 Introduction

Surrogate-based optimization is the standard approach for expensive black-box functions [1], yet many problems contain known, differentiable substructure

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that monolithic surrogates waste samples learning. When a model can be readily separated into an "expensive" black-box component and a "cheap" white-box component, building a single surrogate over all variables forces the surrogate model to learn both the unknown and the known components. The cost of this redundancy grows with dimension: as the size of the combined variable space $[x^{\mathrm{WB}}, x^{\mathrm{BB}}]$ expands, the curse of dimensionality burdens the surrogate for structure it should never have had to discover.

This situation arises naturally in multi-scale engineering problems where design and operation are jointly optimized. For example, a catalyst designer uses density functional theory (DFT) simulations to predict kinetic parameters, which are then used in well-understood reactor mass and energy balances to evaluate reaction yield. Similarly, a polymer engineer uses molecular simulations to estimate membrane transport properties, then solves the solution-diffusion equations to size a separation unit. In each case, the overall objective depends on two groups of decision variables. *Black-box variables* x^{BB} are inputs to expensive simulations or experiments. Conversely, *white-box variables* x^{WB} are variables present in the closed-form macroscopic equation. The black-box and white-box elements of the model are *separable*: the expensive computation depends only on x^{BB}, producing outputs $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$ that parameterize the white-box model. The key structural property is that, for any fixed x^{BB} and y, the remaining optimization over x^{WB} is a standard NLP solvable by conventional methods assuming that the WB equations are continuous and that x^{WB} are all real.

Several lines of work exploit known structure in expensive optimization, ranging from deterministic grey-box solvers [2, 3, 4] to composite and structured Bayesian optimization [5, 6, 7] to data-driven bilevel methods [8, 9]. We review these in Section 2. The common gap is that no prior method combines a surrogate-based global search over only the black-box variables with exact NLP solution of the white-box subproblem—a configuration that fully exploits variable separability for both dimensionality reduction and exact constraint satisfaction.

When the white-box optimization is solvable exactly, the surrogate should operate over $\mathbb{R}^{n_{\mathrm{BB}}}$ rather than $\mathbb{R}^{n_{\mathrm{WB}}+n_{\mathrm{BB}}}$. We exploit this through a bilevel reformulation: the outer loop uses Bayesian Optimization (BO) to search over x^{BB}; for each candidate, we evaluate the black box to obtain intermediate outputs y, then solve the white-box optimization exactly via global optimization to get $x^{\mathrm{WB}\star}$ and $J(x^{\mathrm{WB}\star})$. The GP models only the scalar mapping $x^{\mathrm{BB}} \mapsto J(x^{\mathrm{WB}\star}, y, x^{\mathrm{BB}})$, reducing the surrogate dimension from $n_{\mathrm{WB}} + n_{\mathrm{BB}}$ to n_{BB} and satisfying white-box constraints exactly whenever the inner optimizer converges to a feasible point.

We make three contributions:

1. A bilevel reformulation that reduces surrogate dimensionality from $n_{\mathrm{WB}}+n_{\mathrm{BB}}$ to n_{BB} by solving the white-box subproblem via global optimization, with white-box constraints satisfied exactly whenever the inner optimizer converges to a feasible point—without penalty functions, chance constraints, or moment approximations.

2. A benchmark suite of 13 separable grey-box problems (2–5 total variables, 0–3 constraints) considering synthetic test functions and prototype engineering applications—the largest such suite for this problem class.

3. Comprehensive empirical evidence from 7,410 independent optimization runs (7,280 BO + 130 NLP): the proposed bilevel strategy achieves 14×–277,000× lower regret than black-box BO across all 13 problems, with comparable wall clock time and robustness to hyperparameters (n_{init}, ξ).

2 Related Work

Conceptually, the proposed bilevel framework lies at the intersection of surrogate-based global optimization, grey-box Bayesian optimization, bilevel programming, and constrained expensive optimization. We review each of these areas, focusing on the structural assumptions each approach makes and where those assumptions limit applicability to separable grey-box problems.

2.1 Surrogate-Based Global Optimization

The use of surrogate models to guide expensive optimization traces to the Efficient Global Optimization (EGO) algorithm of Jones et al. [10], which introduced Expected Improvement (EI) as an acquisition function for Gaussian Process (GP) surrogates. EGO treats the objective as a monolithic black box and builds a single GP over the full decision space—an approach that remains the default in most BO implementations [11, 1].

Within the deterministic surrogate realm, Boukouvala, Hasan, and Floudas [2] developed a methodology for constrained grey-box problems that selects surrogates from a diverse set of functional forms and globally optimizes the resulting NLP. The companion ARGONAUT framework [3] formalized this into an iterative algorithm with variable selection, bounds tightening, and constrained sampling for problems with up to 100 variables; Kieslich et al. [12] extended the approach using Smolyak grids and polynomial surrogates. These methods share our philosophy of exploiting known equations, but they surrogate every variable indiscriminately rather than distinguishing black-box from white-box components; the known equations are used only to generate training data or to tighten bounds, not to reduce the surrogate's dimensionality. Moreover, the deterministic surrogates they employ lack the principled exploration–exploitation trade-off of probabilistic models and tend to overexploit early samples under tight evaluation budgets.

The trust-region filter algorithms of Eason and Biegler [4, 13] offer a complementary local approach: they embed surrogates for black-box components within a trust region and solve the white-box NLP at each iteration. This is architecturally close to our inner-loop optimization, but the trust-region framework provides only local convergence guarantees—no global exploration mechanism steers the search away from suboptimal local minima.

2.2 Grey-Box Bayesian Optimization

Grey-box BO methods incorporate known structure into the surrogate to reduce its modeling burden. Astudillo and Frazier [5] introduced Bayesian Optimization of Composite Functions (BOCF) which models composite functions $g(h(x))$ by training a multi-output GP on the expensive inner function h and propagating uncertainty through the known outer function g via sampling; later extensions handle function networks [14] and partial evaluations [15].

Paulson and Lu [6] extend this to constrained grey-box problems by combining multivariate GP models with a constrained expected utility function, using sample average approximation and chance constraints to propagate GP uncertainty through known equations. González and Zavala [16] propose an interconnected-systems BO framework with adaptive linearization, Winz et al. [17] modify the upper confidence bound to exploit grey-box derivative information, and Kudva and Paulson [7] exploit function-network topology for robust optimization.

All of these methods create surrogate models of the *intermediate* black-box outputs y and propagate uncertainty through the white-box equations, requiring multi-output GPs and moment approximations or sampling. When the white-box subproblem is solvable exactly via global optimization—as when $\dim(x^{\mathrm{WB}})$ is moderate—surrogating y adds modeling complexity without clear benefit. We instead surrogate the *optimal value* $J(x^{\mathrm{WB}\star}, y, x^{\mathrm{BB}})$ directly, which yields a scalar GP in $\mathbb{R}^{n_{\mathrm{BB}}}$ and avoids both multi-output surrogates and uncertainty propagation (Table 1).

2.3 Bilevel Optimization

Bilevel programming has a rich history in the global optimization literature. Gümüş and Floudas [18] developed the first rigorous global optimization approach for nonlinear bilevel problems using the αBB framework. Mitsos, Chachuat, and Barton [19] extended this to bilevel dynamic optimization, Kleniati and Adjiman [20] addressed the mixed-integer case, and Faísca et al. [21] proposed parametric global optimization for bilevel programs via multiparametric programming. These exact methods provide optimality guarantees but require analytical access to both levels—an assumption that breaks when the upper level involves an expensive black-box simulation.

Data-driven approaches address this gap. The DOMINO framework of Beykal et al. [8] solves bilevel mixed-integer nonlinear problems by sampling the upper-level objective and solving the lower-level problem to global optimality at each sample point, using a grey-box optimization solver (ARGONAUT/NOMAD) for the upper-level search. DOMINO is the closest non-BO predecessor to our approach: it shares our architecture of exact lower-level solves with a data-driven upper-level search. The key distinction is that DOMINO uses deterministic surrogates for the upper level, whereas we use a GP with a principled acquisition function that balances exploration and exploitation—an advantage when the evaluation budget is small (<250 calls) and the risk of overexploiting early samples is high.

Within the BO literature, Kieffer et al. [9] first applied BO to bilevel problems, using EI at the outer level and Sequential Least Squares Programming (SLSQP) at the inner level. This nested architecture is structurally identical to ours, but Kieffer et al. did not exploit grey-box structure or dimensionality reduction. Recent work addresses the harder setting where *both* levels are black boxes: Ekmekcioglu et al. [22] model both levels as GPs over the joint space; Chew et al. [23] construct confidence-bound trusted sets with regret guarantees; Fu et al. [24] provide convergence analysis for BO bilevel problems where the inner loop is unconstrained Stochastic Gradient Descent (SGD). When the inner problem is a known white-box NLP, solving it exactly is both cheaper and more accurate than learning a second surrogate.

Baldea [25] introduced the multi-scale bilevel BO framework we extend here, demonstrating it on a single chemical process case study. We advance that work with a 13-problem benchmark suite, with systematic hyperparameter sweeps, and detailed comparisons quantifying when and why bilevel decomposition helps.

2.4 Constraint Handling in Expensive Optimization

Standard constrained BO methods model constraints as expensive black-box functions, each requiring a separate GP. Gardner et al. [26] introduced weighted expected improvement for unknown constraints; Gelbart et al. [27] extended this with probability-of-feasibility weighting; Picheny et al. [28] proposed a slack-variable augmented Lagrangian approach. These methods are appropriate when constraint functions are truly unknown, but when constraints arise from known white-box equations—mass balances, safety envelopes, thermodynamic limits—creating surrogate models increases computational cost and introduces approximation error.

Our inner global optimizer satisfies white-box constraints exactly, avoiding penalty functions, chance constraints, and moment approximations. The advantage is largest when feasible regions are tight: in such cases(reflected in the Distillation and Heat-Exchanger problems in Section 6), bilevel BO achieves $>$100,000$\times$ lower regret than penalty-based black-box BO.

3 Problem Formulation

We consider optimization problems in which an expensive black-box model is coupled with known, differentiable equations. This setting arises in multi-scale engineering design—molecular simulations feeding into process models—but also in any domain where part of the system is analytically tractable and part is not. We first formalize the general problem, then identify the structural property our method exploits.

3.1 Problem Setting and Notation

The decision variables comprise two groups:

- $x^{\mathrm{WB}} \in \mathbb{R}^{n_{\mathrm{WB}}}$: *white-box variables* governed by known, differentiable equations (e.g., temperatures, pressures, flow rates in engineering; policy parameters in control; design dimensions in structural optimization). These variables enter the white-box model f^{WB}, the objective J, and the inequality constraints g.

- $x^{\mathrm{BB}} \in \mathbb{R}^{n_{\mathrm{BB}}}$: *black-box variables* that parameterize an expensive, non-differentiable function f^{BB} (e.g., molecular descriptors requiring DFT, material properties from simulations, hyperparameters of a costly simulator). These variables enter f^{BB} directly and may also appear in J and g.

The black-box model

$$y = f^{\mathrm{BB}}(x^{\mathrm{BB}}), \quad f^{\mathrm{BB}} : \mathbb{R}^{n_{\mathrm{BB}}} \to \mathbb{R}^{n_y}, \tag{1}$$

maps x^{BB} to a vector of intermediate parameters $y \in \mathbb{R}^{n_y}$ that couple the black-box and white-box components. We assume f^{BB} is expensive to evaluate and that we lack closed-form expressions or derivative information for it.

The white-box model takes the implicit form

$$f^{\mathrm{WB}}(x^{\mathrm{WB}}, y) = 0, \quad f^{\mathrm{WB}} : \mathbb{R}^{n_{\mathrm{WB}}} \times \mathbb{R}^{n_y} \to \mathbb{R}^{n_{\mathrm{WB}}}, \tag{2}$$

encoding known equations (e.g., mass and energy balances) that relate the white-box variables x^{WB} to the black-box outputs y. We assume f^{WB} is sufficiently smooth and that the Jacobian $\partial f^{\mathrm{WB}}/\partial x^{\mathrm{WB}}$ is available analytically.

The inequality constraints

$$g(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) \leq 0, \quad g : \mathbb{R}^{n_{\mathrm{WB}}} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{\mathrm{BB}}} \to \mathbb{R}^{n_g}, \tag{3}$$

encode n_g design specifications, safety limits, or feasibility requirements that may depend on both variable groups and the black-box outputs.

The complete optimization problem over both variable groups is

$$\begin{aligned}
\min_{x^{\mathrm{WB}}, x^{\mathrm{BB}}} \quad & J(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) && \text{(4a)}\\
\text{s.t.} \quad & f^{\mathrm{WB}}(x^{\mathrm{WB}}, y) = 0 && \text{(4b)}\\
& y = f^{\mathrm{BB}}(x^{\mathrm{BB}}) && \text{(4c)}\\
& g(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) \leq 0 && \text{(4d)}\\
& x^{\mathrm{WB}} \in \mathcal{X}^{\mathrm{WB}}, \quad x^{\mathrm{BB}} \in \mathcal{X}^{\mathrm{BB}} && \text{(4e)}
\end{aligned}$$

The objective $J(x^{\mathrm{WB}}, y, x^{\mathrm{BB}})$ and inequality constraints $g(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) \leq 0$ may depend on the white-box variables, the black-box outputs, and—in some problems—directly on the black-box variables x^{BB}. The bilevel decomposition remains valid because, for any fixed x^{BB} and $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$, the inner optimization is over x^{WB} alone. The sets $\mathcal{X}^{\mathrm{WB}}$ and $\mathcal{X}^{\mathrm{BB}}$ define box constraints.

3.2 Separable Grey-Box Structure

Problem (4) possesses a structural property that we formalize as an assumption and then exploit algorithmically.

Assumption 1 (Separable grey-box structure)**.** Problem (4) satisfies:

(i) The black-box function f^{BB} depends only on x^{BB}. For any fixed x^{BB} and $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$, the resulting optimization over x^{WB} alone is a well-posed NLP. (The objective J and constraints g may depend on x^{BB} directly, not only through y; the separability is in the *optimization structure*—the inner problem optimizes x^{WB} while x^{BB} and y are held fixed.)

(ii) For any fixed x^{BB} and $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$, the white-box subproblem $\min_{x^{\mathrm{WB}}} J(x^{\mathrm{WB}}, y, x^{\mathrm{BB}})$ subject to $f^{\mathrm{WB}}(x^{\mathrm{WB}}, y) = 0$, $g(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) \leq 0$, $x^{\mathrm{WB}} \in \mathcal{X}^{\mathrm{WB}}$ is solvable to global optimality (of the subproblem) by a suitable global optimizer.

Under Assumption 1, the optimization can be decomposed as follows: for any fixed x^{BB}, the black-box evaluation $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$ and the inner optimization over x^{WB} are self-contained. This decoupling is the structural feature we exploit: rather than optimizing all variables jointly, we can search over the low-dimensional black-box space while solving the white-box subproblem exactly at each step. Note that f^{BB} can always be augmented with identity mappings (e.g., $y_{n_y+1} = x_1^{\mathrm{BB}}$) to absorb any direct x^{BB} dependence in J or g, so the distinction between "depends on x^{BB} through y" and "depends on x^{BB} directly" is a modeling choice, not a structural limitation.

Despite this structure, problem (4) resists standard solution methods for two reasons. First, the black-box nature of f^{BB} precludes using gradient-based NLP solvers. Finite-difference gradient approximations require $O(n_{\mathrm{BB}})$ evaluations of f^{BB} per iteration—prohibitive when each evaluation takes minutes to hours. Second, full-space surrogate-based optimization (EGO [10], standard BO [1]) over $[x^{\mathrm{WB}}, x^{\mathrm{BB}}]$ builds a GP surrogate in $\mathbb{R}^{n_{\mathrm{WB}}+n_{\mathrm{BB}}}$, wasting samples on the known white-box response and suffering the curse of dimensionality as n_{WB} grows.

4 Method: Bilevel Bayesian Optimization

4.1 Bilevel Reformulation

Based on the assumptions in Section 3.2 we reformulate (4) as a bi-level optimization problem that separates what we know from what we must learn:

$$
\begin{aligned}
\min_{x^{\mathrm{BB}}} \quad & J(x^{\mathrm{WB}\star}, y, x^{\mathrm{BB}}) && \text{(5a)}\\
\text{s.t.} \quad & y = f^{\mathrm{BB}}(x^{\mathrm{BB}}) && \text{(5b)}\\
& x^{\mathrm{WB}\star} = \arg\min_{x^{\mathrm{WB}}} \; J(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) && \text{(5c)}\\
& \qquad \text{s.t.} \quad f^{\mathrm{WB}}(x^{\mathrm{WB}}, y) = 0 && \text{(5d)}\\
& \qquad \qquad g(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) \le 0 && \text{(5e)}\\
& \qquad \qquad x^{\mathrm{WB}} \in \mathcal{X}^{\mathrm{WB}} && \text{(5f)}\\
& x^{\mathrm{BB}} \in \mathcal{X}^{\mathrm{BB}} && \text{(5g)}
\end{aligned}
$$

The **inner problem** (5c)–(5f) is a standard nonlinear program (NLP). Given fixed x^{BB} and $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$, it optimizes only the white-box variables x^{WB} using the white box model; x^{BB} and y enter as parameters. Because the white-box equations are closed-form and inexpensive to evaluate, this subproblem can be solved to global optimality using a global optimizer such as differential evolution (DE).

The **outer problem** (5a)–(5b) searches over black-box decisions x^{BB}. For each candidate x^{BB}, we evaluate the black box to obtain y, solve the inner problem to global optimality to get $x^{\mathrm{WB}\star}$, and record the resulting objective value $J(x^{\mathrm{WB}\star}, y, x^{\mathrm{BB}})$. This outer loop is where we apply Bayesian optimization.

Remark 1. The key advantage is dimensionality reduction. The BO surrogate now operates over $\mathbb{R}^{n_{BB}}$ rather than $\mathbb{R}^{n_{WB}+n_{BB}}$. When $n_{BB} \ll n_{WB}$, the number of black-box evaluations required to build an accurate surrogate drops substantially relative to applying BO to problem (4).

Remark 2. If the inner problem (5c)–(5f) is infeasible for some y, we return a large penalty value. The BO acquisition function will naturally steer away from black-box variable choices that lead to infeasible white-box solutions.

4.2 Algorithm, Surrogate Model, and Acquisition Function

Algorithm 1 Bilevel Bayesian Optimization

Require: Black-box bounds $\mathcal{X}^{\mathrm{BB}}$, white-box bounds $\mathcal{X}^{\mathrm{WB}}$, models f^{BB}, f^{WB}, J, g; initial samples n_{init}, iterations N, exploration parameter ξ
Ensure: Best $x^{\mathrm{BB}\star}$, $x^{\mathrm{WB}\star}$, $J^\star$

// Initialization
Sample $\{x_i^{\mathrm{BB}}\}_{i=1}^{n_{\mathrm{init}}}$ via Latin hypercube in $\mathcal{X}^{\mathrm{BB}}$
for $i = 1, \ldots, n_{\mathrm{init}}$ **do**
    Evaluate $y_i = f^{\mathrm{BB}}(x_i^{\mathrm{BB}})$
    Solve inner problem (DE): $x_i^{\mathrm{WB}\star} = \arg\min_{x^{\mathrm{WB}}} J(x^{\mathrm{WB}}, y_i, x_i^{\mathrm{BB}})$ s.t. $f^{\mathrm{WB}}(x^{\mathrm{WB}}, y_i) = 0$, $g(x^{\mathrm{WB}}, y_i, x_i^{\mathrm{BB}}) \leq 0$
    Record $J_i = J(x_i^{\mathrm{WB}\star})$ (or penalty if infeasible)
end for
// BO loop
for $t = n_{\mathrm{init}} + 1, \ldots, n_{\mathrm{init}} + N$ **do**
    Fit GP surrogate $\mathcal{GP}$ to $\{(x_i^{\mathrm{BB}}, J_i)\}_{i=1}^{t-1}$
    Select $x_t^{\mathrm{BB}} = \arg\max_{x^{\mathrm{BB}} \in \mathcal{X}^{\mathrm{BB}}} \alpha_{\mathrm{EI}}(x^{\mathrm{BB}}; \mathcal{GP}, \xi)$
    Evaluate $y_t = f^{\mathrm{BB}}(x_t^{\mathrm{BB}})$
    Solve inner problem (DE) $\to x_t^{\mathrm{WB}\star}$, record J_t
    Update dataset
end for
return best $(x^{\mathrm{BB}\star}, x^{\mathrm{WB}\star}, J^\star)$

We describe the proposed approach in Algorithm 1 and Figure 1 and provide a discussion as follows: The GP surrogate in Algorithm 1 models the scalar mapping $J(x^{\mathrm{WB}\star}(x^{\mathrm{BB}}), f^{\mathrm{BB}}(x^{\mathrm{BB}}), x^{\mathrm{BB}}) : \mathbb{R}^{n_{\mathrm{BB}}} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{\mathrm{WB}}} \mapsto \mathbb{R}$—a function of x^{BB} alone. In our implementation, we use a Gaussian process with an automatic relevance determination (ARD) radial basis function (RBF) kernel, where a separate length scale per dimension allows the GP to adapt to anisotropic landscapes. The outer loop selects the next evaluation by maximizing Expected Improvement (EI) with exploration parameter $\xi \geq 0$. Kernel and acquisition function details for our implementation are in Appendices C and B; here we emphasize that both are standard—the gains reported in Section 6 come from problem structure, not from novel surrogate or acquisition choices, although we expect that adopting them would yield further benefits.

4.3 Key Properties

The bilevel reformulation (5) has several properties worth highlighting. The first is formalized as a proposition.

Proposition 1. *Under Assumption 1, the bilevel reformulation* (5) *preserves the global optimum of* (4) *and reduces the GP surrogate domain from* $\mathbb{R}^{n_{\mathrm{WB}}+n_{\mathrm{BB}}}$ *to* $\mathbb{R}^{n_{\mathrm{BB}}}$.

The proof is in Appendix F. The key consequences are threefold:

Exact constraint satisfaction. Constraints $g(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) \leq 0$ are handled by the inner global optimizer, not by penalty functions or chance constraints.

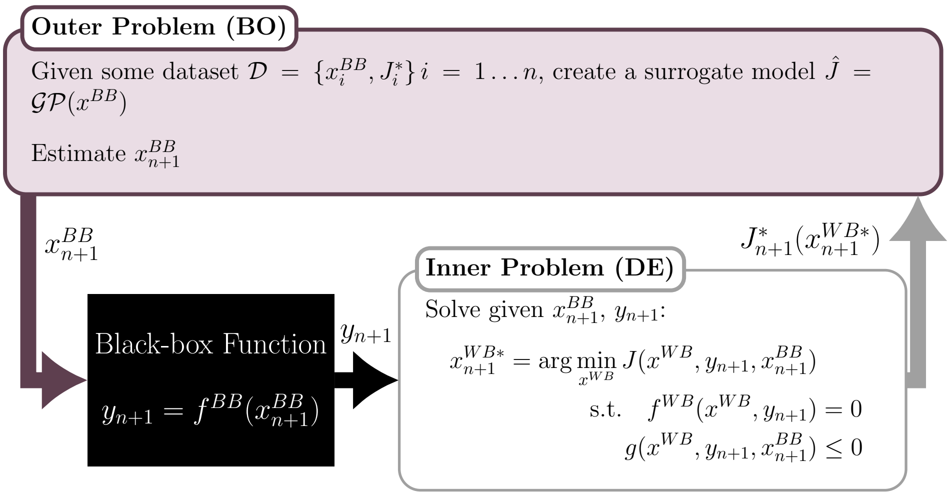

Figure 1: Flow of the multi-scale Bayesian optimization framework. The outer loop uses BO to select black-box variables x^{BB}, which are evaluated through the black-box model to produce parameters y. The inner loop solves the white-box optimization over x^{WB} via differential evolution given x^{BB} and y, returning the optimal objective $J(x^{\mathrm{WB}\star}, y, x^{\mathrm{BB}})$ to update the BO surrogate.

The GP never needs to learn the constraint boundary. When the inner optimizer converges to a globally feasible point (Assumption 1(ii)), feasibility is guaranteed. In our implementation, differential evolution with L-BFGS-B polishing provides high-confidence global solutions; further guarantees could be obtained with deterministic global solvers (e.g., BARON), as discussed in Section 7.

Scalar surrogate. The method requires only one GP for the scalar optimal objective $J(x^{\mathrm{WB}\star}, y, x^{\mathrm{BB}})$. Approaches that surrogate the black-box outputs $y \in \mathbb{R}^{n_y}$ instead require a multi-output GP, with the associated challenges of cross-covariance specification and $O(n_y^3)$ scaling.

Additional implementation details—infeasibility handling and solver modularity—are discussed in Appendix G.

4.4 Relationship to Existing Grey-Box Methods

Table 1: Comparison of grey-box BO approaches. Bilevel BO (this work) is the only method that both separates variables and solves the white-box subproblem exactly.

	Bilevel BO (this work)	**COBALT** [6]	**BOCF** [5]	**Full-space BO**
What is surrogated	$x^{\mathrm{BB}} \mapsto J^\star$	$x^{\mathrm{BB}} \mapsto y$	$x \mapsto h(x)$	$x \mapsto J$
Surrogate type	scalar GP	multi-output GP	vector GP	scalar GP
Surrogate dimension	n_{BB}	n_{BB}	$n_{\mathrm{WB}} + n_{\mathrm{BB}}$	$n_{\mathrm{WB}} + n_{\mathrm{BB}}$
Variables separated?	Yes	Yes	No	No
Constraint handling	exact (global opt.)	chance constraints	penalty	penalty
Inner optimization	global optimization (DE)	moment propagation	none	none
y-dependent constraints	via inner optimizer	natively	penalty	penalty

Table 1 summarizes the distinctions between the proposed method and a subset of the methods reviwed in Section 2 that are most directly related and relevant. Both bilevel BO and COBALT achieve the same input dimensionality reduction to n_{BB}. The distinction is in the surrogate output (scalar vs. multi-output GP) and constraint handling (exact global optimization vs. moment approximation), not the input dimension. COBALT and BOCF surrogate intermediate black-box outputs y and propagate uncertainty through the white-box equations, requiring multi-output GPs and moment approximations. Our method surrogates the scalar optimal objective $J(x^{\mathrm{WB}\star})$ directly, at the cost of an inner global optimization solve per outer iteration. When the white-box model is moderately sized and the black-box evaluation is the primary computational expense, this overhead is negligible. The approaches are complementary: COBALT handles y-dependent constraints natively and enables risk-averse decisions via uncertainty propagation; BOCF handles non-separable composite functions. Our method trades their generality for simplicity and exactness when Assumption 1 holds.

5 Benchmark Problem Suite

We propose a suite of 13 problems spanning synthetic test functions and engineering domains, summarized in Table 2. Full mathematical formulations are in Appendix A.The suite is designed around five principles: (1) every problem has a verified global optimum (Appendix E); (2) constraints are physically motivated; (3) black-box functions encode realistic structure-property relationships (Sabatier volcano curves, Langmuir isotherms, Robeson upper bounds, Hansen solubility); (4) problems span a range of difficulty (unconstrained to 3 constraints, 2D to 5D, min and max); and (5) all problems are cheap to evaluate, enabling the 10,920-run statistical comparison in Section 6.

To our knowledge, no existing benchmark suite targets a separable grey-box structure that reflects Assumption 1: an outer black-box mapping $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$ coupled with an inner white-box optimization over x^{WB}, subject to inequality constraints. Existing problem suites for composite BO [5] do not separate variables; those used in grey-box BO [6] contain only a handful of problems; and standard global optimization testbeds such as the IEEE Congress on Evolutionary Computation (CEC) benchmarks [29] and the Comparing Continuous Optimizers (COCO) platform with its Black-Box Optimization Benchmarking (BBOB) suite [30] lack bilevel structure entirely.

5.1 Representative Problems

We highlight two problems; full formulations for all 13 appear in Appendix A. Adapted from Gardner et al. [26] and Ariafar et al. [31], these 2D constrained minimization problems feature small, disconnected feasible regions. The two-dimensional structure permits direct visualization of the search behavior (Section 6.2).

Table 2: Summary of the 13-problem benchmark suite. n_{WB}: white-box variables, n_{BB}: black-box variables, n_y: black-box outputs, n_g: inequality constraints.

Problem	n_{WB}	n_{BB}	n_y	n_g	**Type**	**Domain**
Small-Feasible-Region 1	1	1	1	1	min	Synthetic
Small-Feasible-Region 2	1	1	1	1	min	Synthetic
Rastrigin	2	1	1	0	min	Multimodal
Toy-Hydrology	1	1	1	2	min	Hydrology
Rosen-Suzuki	2	2	2	3	min	Constrained
CSTR	3	2	3	2	max	Catalysis
Heat-Exchanger	3	2	3	2	min	Heat Exchanger Network design
PSA	3	2	3	2	min	Adsorption
Batch-Reactor	3	2	3	2	min	Kinetics
Distillation	3	2	3	3	min	Separation
Evaporator	3	2	3	2	min	Evaporation
Membrane	3	2	3	2	min	Membrane sep.
Williams-Otto	3	2	2	2	max	Process opt.

6 Computational Experiments

6.1 Experimental Design

6.1.1 Solvers Compared

We compare three solvers that represent different ways of handling the grey-box structure in problem (4).

The first solver, **black-box DE**, applies SciPy's differential evolution (DE) solver over the full variable space $[x^{\mathrm{WB}}, x^{\mathrm{BB}}]$ with a population size multiplier of 15 and L-BFGS-B polishing. Constraints are handled via `NonlinearConstraint` with $g \leq 0$. This solver provides a global-search baseline: DE is gradient-free and explores the full feasible region without requiring derivative access to the black-box model. However, it evaluates f^{BB} at every candidate in the population at every generation, so the total number of black-box evaluations grows rapidly with dimension.

The second solver, **black-box BO (BB-BO)**, applies standard Bayesian optimization over the full $(n_{\mathrm{WB}} + n_{\mathrm{BB}})$-dimensional space. A single GP surrogate models the mapping $[x^{\mathrm{WB}}, x^{\mathrm{BB}}] \mapsto J$. Constraints are handled via a penalty function. This solver ignores the separable structure entirely and serves as the BO baseline.

The third solver, **bilevel BO (Bi-BO)**, is the proposed method. It places the GP surrogate over only the n_{BB}-dimensional black-box space and solves the constrained white-box subproblem globally using DE, as described in Algorithm 1.

All three solvers use Expected Improvement (EI) as the acquisition function for the BO variants. Both BO solvers share the same ARD RBF kernel and GP configuration (Appendix C). We sweep over initial sample sizes $n_{\mathrm{init}} \in \{1, 5, 20, 50\}$ and exploration parameters $\xi \in \{0.001, \ldots, 1.0\}$, yielding 7,410 independent runs across all 13 problems (Appendix I).

6.1.2 Metrics

We report four metrics. *Simple regret* $r_t = |J_{\text{best}}(t) - J^\star|$ measures the gap between the best objective found after t evaluations and the verified global optimum $J^\star$. *Convergence speed* is the number of iterations needed to reduce regret to 1% of its initial value. *Wall time* is the total elapsed time for $n_{\text{init}}+200$ evaluations. Finally, we count the total number of *black-box evaluations* (calls to f^{BB}) to assess sample efficiency.

6.2 Mechanism Visualization

We begin with the two Small-Feasible-Region problems because their two-dimensional structure permits direct visualization of how bilevel BO searches differently from black-box BO. Both problems have one white-box variable (x^{WB}) and one black-box variable (x^{BB}), with small, disconnected feasible regions defined by nonlinear constraints.

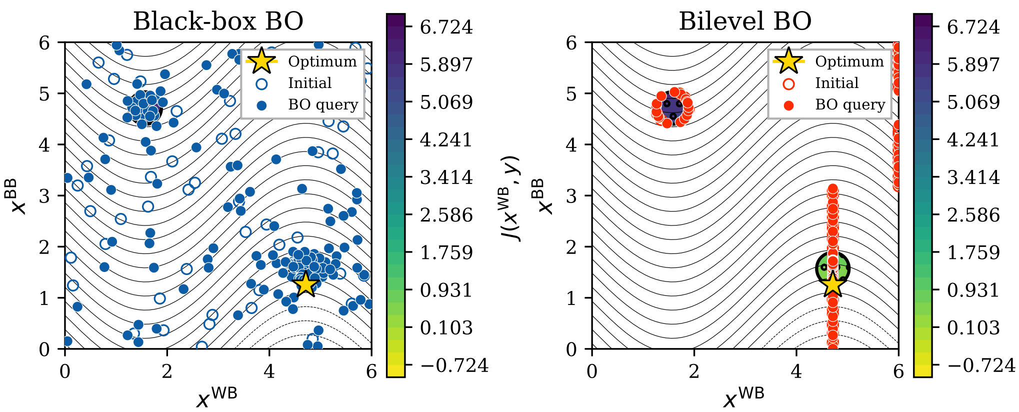

Figure 2: Small-Feasible-Region 1: search behavior of black-box BO (left) and bilevel BO (right) after 50 initial samples + 200 BO iterations. Background: objective $J(x^{\text{WB}}, y)$ (yellow = low); white regions are infeasible; hatching marks feasible islands; $\star$ marks the global optimum. Black-box BO scatters queries across the full $(x^{\text{WB}}, x^{\text{BB}})$ space. Bilevel BO operates along the x^{BB} axis—each query selects an x^{BB} value, and the inner DE solver finds the best feasible x^{WB}.

Figure 2 shows the key difference. In SFR-1, the black-box function is the identity ($y = x^{\text{BB}}$), so the bilevel decomposition reduces to: given a candidate y-value from the outer loop, find the x^{WB} that minimizes $\sin(x^{\text{WB}}) + y$ within the feasible set. The inner DE solver handles constraint satisfaction exactly, so every bilevel BO query lands in a feasible region. Black-box BO, by contrast, must jointly learn the objective function landscape *and* the constraint boundary in two dimensions, wasting many samples in infeasible regions.

SFR-2 (Figure 3) introduces a nonlinear black-box function, making the mapping from x^{BB} to the feasible objective landscape more complex. The same mechanism applies: the outer loop proposes x^{BB}, the black box function is used to compute y, and the inner DE solver finds the best feasible x^{WB}. The GP surrogate now models a one-dimensional function $x^{\text{BB}} \mapsto J(x^{\text{WB}\star})$ rather than the two-dimensional objective-plus-constraint landscape that black-box BO must learn.

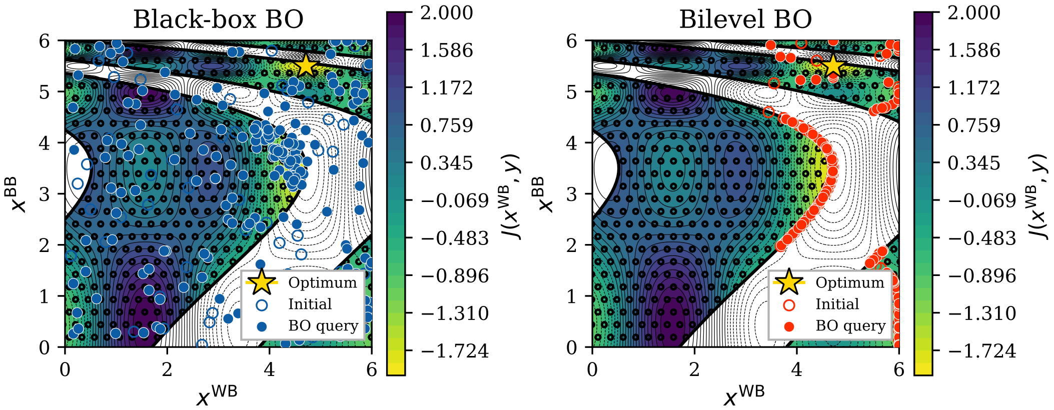

Figure 3: Small-Feasible-Region 2: same layout as Figure 2. Here the black-box function is nonlinear ($y = \frac{x^{\mathrm{BB}}-5.5}{2}\exp(x^{\mathrm{BB}}/2)$), distorting the relationship between x^{BB} and the feasible set. Bilevel BO still operates near the optimum because the GP surrogate learns the scalar mapping $x^{\mathrm{BB}} \mapsto J(x^{\mathrm{WB}\star})$ in one dimension.

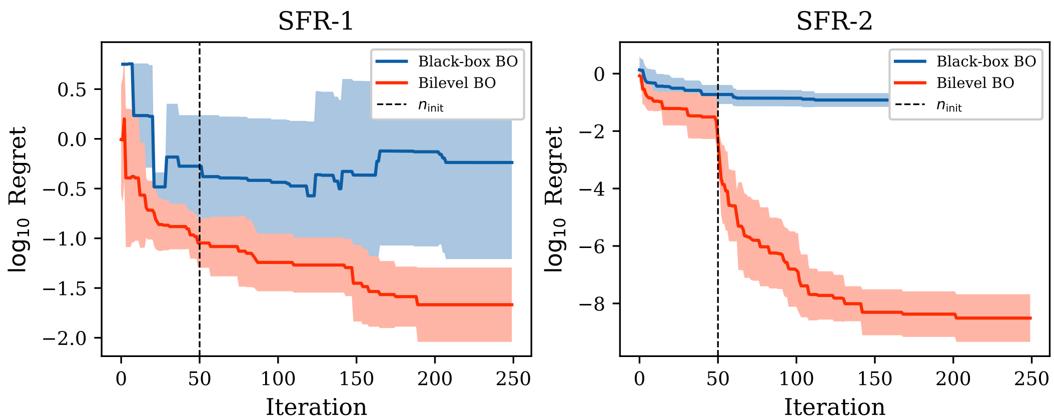

Figure 4: Regret convergence for the two Small-Feasible-Region problems. Each curve uses $n_{\mathrm{init}} = 50$ initial Latin hypercube samples and the EI exploration parameter ξ that yielded the lowest final regret. The dashed line marks the end of initialization. Solid lines show the mean over 10 independent repetitions; shaded bands show ± 1 standard deviation. Bilevel BO (red) converges 1–2 orders of magnitude below black-box BO (blue) on both problems.

The convergence curves (Figure 4) confirm the visual impression. On SFR-1, bilevel BO reduces regret to $\sim 10^{-1.5}$ while black-box BO plateaus near $10^{0.4}$—a 76× improvement. On SFR-2, bilevel BO reaches the global optimum (regret $< 10^{-4}$) while black-box BO stalls at $10^{-0.9}$—a 42,743× improvement. In both cases, the mechanism is the same: bilevel BO needs only to learn a one-dimensional function, while black-box BO must simultaneously model a two-dimensional objective and discover the small feasible islands.

6.3 Main Results

The pattern observed on the SFR problems extends to all 13 benchmarks. Table 3 reports final regret after 200 BO iterations; Figure 5 shows convergence curves for the remaining 11 problems.

Table 3: Final regret after 200 iterations ($n_{\text{init}} = 50$, best ξ; see Appendix I for per-problem ξ values). BB/Bi ratio quantifies the advantage of bilevel BO. Bilevel BO achieves lower regret on every problem.

Problem	**dim**	**NLP**	**BB-BO**	**Bi-BO**	**BB/Bi**
Small-Feasible-Region	2	0.0000	3.2002	0.0420	76×
Small-Feasible-Region-2	2	0.0546	0.1798	0.0000	42,743×
Rastrigin	3	3.3829	7.9551	0.0001	57,717×
Toy-Hydrology	2	0.0000	0.1278	0.0000	6,459×
Rosen-Suzuki	4	0.0000	6.8729	0.4763	14×
CSTR	5	0.0000	2.8773	0.0052	552×
Heat-Exchanger	5	0.0002	51.1785	0.0002	276,962×
PSA	5	0.0000	0.4743	0.0046	103×
Batch-Reactor	5	0.0000	1.5285	0.0354	43×
Distillation	5	0.0000	5556.2612	0.0321	173,088×
Evaporator	5	0.0000	0.7427	0.0393	19×
Membrane	5	0.0000	16.5371	0.0001	115,095×
Williams-Otto	5	0.0000	2.2550	0.0000	115,769×

Bilevel BO achieves 14×–277,000× lower regret than black-box BO across all 13 problems. The gap opens early and widens: bilevel BO separates from black-box BO within 20–50 iterations on most problems. The largest gains (Heat-Exchanger: 277,000×, Distillation: 173,000×) occur where tight constraints hinder the penalty-based approach—the same mechanism illustrated by the SFR figures, but now in five dimensions where the advantage of exact constraint satisfaction compounds with dimensionality reduction. The smallest gains (Rosen-Suzuki: 14×, Evaporator: 19×) occur where the inner white-box subproblem—solved via differential evolution with L-BFGS-B polishing—has multiple local optima that prevent the solver from reliably reaching the global optimum, or where the constraint set is loose.

The full-space DE baseline—which applies differential evolution over the joint $[x^{\text{WB}}, x^{\text{BB}}]$ space with population size 15 and L-BFGS-B polishing—fails on two problems: Rastrigin, where multimodality in the combined space challenges even global solvers, and Small-Feasible-Region-2, where the narrow feasible region limits convergence. Bilevel BO matches or approaches DE baseline

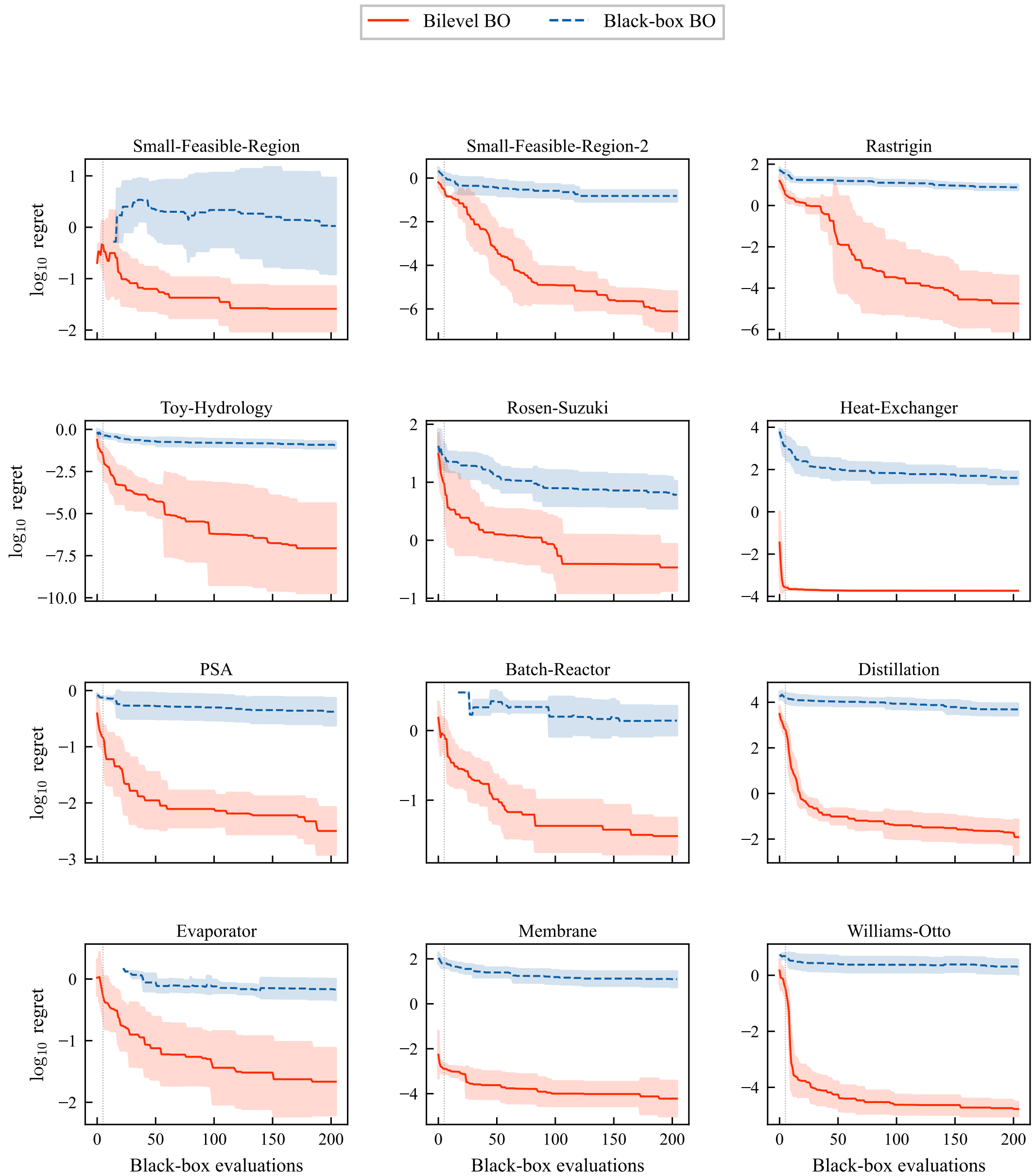

Figure 5: Regret convergence for the 11 benchmark problems not shown in Figure 4. Bilevel BO (red) converges to values that are 1–4 orders of magnitude below black-box BO (blue) on every problem. Each curve uses $n_{\text{init}} = 50$ initial Latin hypercube samples and the EI exploration parameter ξ that yielded the lowest final regret for that problem (see Appendix I for per-problem values). Solid lines show the mean over 10 independent repetitions; shaded bands show ± 1 standard deviation.

quality on the remaining 11 problems using ~250 black-box evaluations versus the substantially higher evaluation counts required by full-space DE. All 13 pairwise comparisons (bilevel vs. black-box BO) are statistically significant ($p < 0.01$, Wilcoxon signed-rank test, 10 repetitions).

6.4 Convergence Speed

Where both methods reach 1% of initial regret, bilevel BO converges 1.5×–22× faster (Appendix Table 15). The most striking case is Williams-Otto, where bilevel BO converges in 9 iterations (~14 total evaluations) while black-box BO never reaches the target. Several problems do not reach the 1% threshold within 200 iterations for either method, yet bilevel BO still achieves orders-of-magnitude lower final regret (Table 3).

6.5 Scaling and Robustness

To verify that the bilevel advantage is not an artifact of a particular hyperparameter setting, Figure 6 plots final regret for every (problem, n_{init}, ξ) configuration. Every point lies below the diagonal, confirming that bilevel BO significantly outperforms black-box BO across all 364 configurations tested.

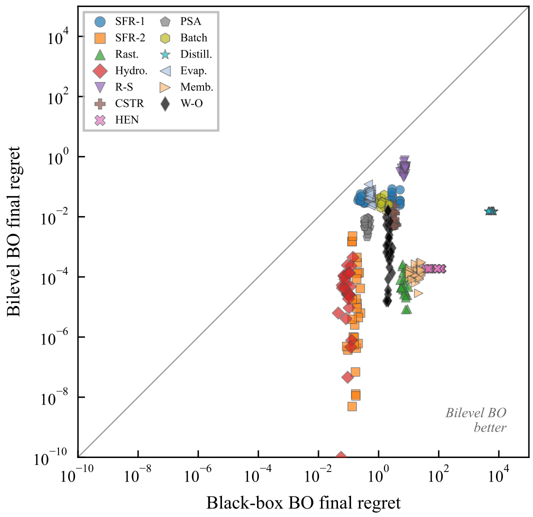

Figure 6: Scatter plot of final regret: black-box BO vs. bilevel BO. Each point is one (problem, n_{init}, ξ) configuration. All points lie below the diagonal, confirming that bilevel BO significantly outperforms BB-BO regardless of hyperparameter choice.

6.5.1 Hyperparameter Sensitivity

Figure 7 examines sensitivity to the number of initial samples. Bilevel BO performance is nearly flat across $n_{\text{init}} \in \{1, 5, 20, 50\}$ (numerical values in Appendix Table 16): even $n_{\text{init}} = 1$—a single initial sample—performs comparably to $n_{\text{init}} = 50$, suggesting that few expensive initial evaluations suffice when f^{BB} is costly.

The bilevel advantage is similarly robust to the EI exploration parameter ξ. No single ξ is universally optimal—the best setting ranges from $\xi =$

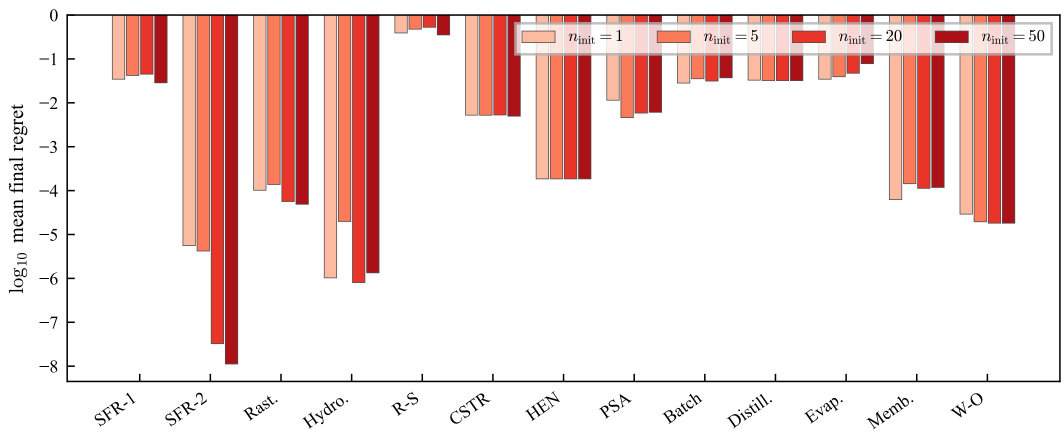

Figure 7: Sensitivity to n_{init}. Final regret ($\log_{10}$ scale) of bilevel BO is nearly flat across $n_{\text{init}} \in \{1, 5, 20, 50\}$, using the EI exploration parameter ξ that yielded the lowest final regret for each problem. Even $n_{\text{init}} = 1$ performs comparably to $n_{\text{init}} = 50$.

0.001 (CSTR, Distillation, Williams-Otto) to $\xi = 0.5$ (Small-Feasible-Region-1, Membrane)—but bilevel BO significantly outperforms black-box BO at every ξ tested, and adaptive ξ strategies are not critical (Appendix Figure 11).

6.5.2 Computational Cost

Both BO methods require ~20–30s wall time for 200 iterations on 12 of 13 problems, as the lower-dimensional GP in bilevel BO roughly offsets the inner DE cost. The exception is Williams-Otto (111s vs. 26s), where the inner DE solves a 6-species steady-state mass balance at each iteration. Both BO methods use ~205 black-box evaluations—an order of magnitude fewer than those required by the full-space DE baseline.

7 Discussion

Bilevel BO achieves 14×–277,000× lower regret than black-box BO on every problem tested, with comparable computational cost. The source of this advantage is structural, not algorithmic: we use the same GP kernel, the same acquisition function, and the same optimizer as the black-box baseline. The gains come entirely from exploiting the separable problem structure—reducing the surrogate dimension and satisfying the white box constraints exactly. Three additional analyses—feasibility rates, a two-factor decomposition of the advantage, and sample efficiency crossover—provide mechanistic insight into when and why the bilevel decomposition helps.

Constraint satisfaction and wasted evaluations. A key mechanism underlying the regret gap is that black-box BO wastes evaluations on infeasible queries. Figure 8 reports the fraction of all 250 evaluations that land in feasible regions, reconstructed from stored search histories. Black-box BO wastes 6–72% of its budget on infeasible points depending on the problem, with Small-Feasible-Region (72%), Batch-Reactor (71%), and Evaporator (58%) being the worst cases. Bilevel BO achieves 100% feasibility on 8 of 12 constrained problems because the inner DE solver handles constraint satisfaction exactly.

On the remaining four problems (Batch-Reactor, Evaporator, Small-Feasible-Region, PSA), bilevel BO feasibility ranges from 59–91%. This occurs for two reasons: (1) the inner DE may return solutions where the total constraint violation $\max_i g_i(x^{\mathrm{WB}}, y)$ is positive but small (on the order of 10^{-4}–10^{-2}), placing the solution near but outside the feasible boundary; or (2) certain $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$ values yield an inner problem with no feasible solution, because no $x^{\mathrm{WB}} \in \mathcal{X}^{\mathrm{WB}}$ satisfies $g(x^{\mathrm{WB}}, y) \leq 0$. This is not a limitation of the bilevel architecture itself but of the inner solver; deterministic global solvers (e.g., BARON) would restore the 100% guarantee.

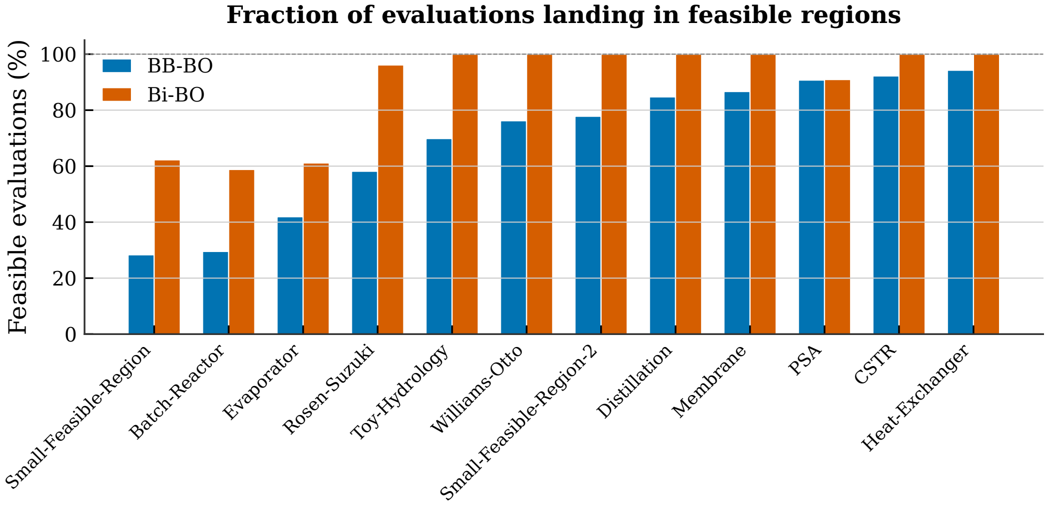

Figure 8: Fraction of evaluations landing in feasible regions across 12 constrained problems ($n_{\mathrm{init}} = 50$, best ξ, mean over 10 repetitions). Black-box BO (blue) wastes 6–72% of its budget on infeasible queries; bilevel BO (red) achieves $\geq$59% feasibility on all problems and 100% on 8 of 12.

Disentangling dimensionality reduction from constraint satisfaction. The regret ratios in Table 3 vary by four orders of magnitude across problems of the same total dimension. Figure 9 decomposes this variation by plotting total dimension against the feasible fraction of the domain (estimated by 10^5 uniform random samples), with the regret ratio encoded as color. Two patterns emerge. First, dimensionality reduction is the dominant factor: Rastrigin (unconstrained, 100% feasible domain) achieves a 57,717$\times$ improvement (at $n_{\mathrm{init}} = 50$, $\xi = 0.2$; Table 3) purely from reducing the GP surrogate from 3D to 1D. Second, constraint tightness modulates but does not determine the gain: Evaporator and Batch-Reactor have the *tightest* constraints (<1.1% feasible domain) yet the *smallest* regret ratios (6$\times$ and 41$\times$), because their nonconvex inner landscapes cause the inner DE to find local optima. Conversely, Heat-Exchanger has a loose constraint set (93% feasible) but the largest gain (277,000$\times$), driven entirely by dimensionality reduction from 5D to 2D. The analysis confirms that when the inner optimizer is well-behaved, dimensionality reduction alone produces large gains; exact constraint satisfaction provides additional benefit but is neither necessary nor sufficient.

Sample efficiency. Beyond final regret, practitioners are interested in how quickly a method reaches useful solutions. Figure 10 reports the number of evaluations at which bilevel BO first matches the final regret that black-box BO

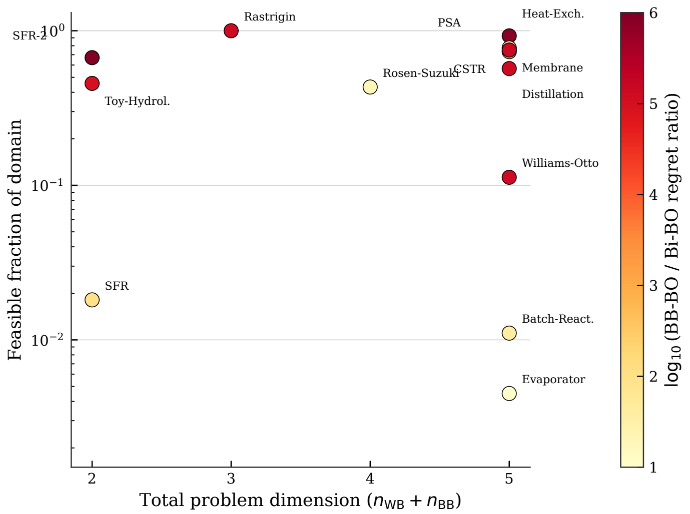

Figure 9: Regret ratio decomposed by total dimension and constraint tightness. Each point is one benchmark problem; color encodes $\log_{10}$(BB-BO / Bi-BO regret ratio). The feasible fraction of the domain is estimated by 10^5 uniform random samples. Dimensionality reduction is the dominant factor; constraint tightness modulates but does not determine the advantage.

achieves after all 250 evaluations. On 5 of 13 problems (Heat-Exchanger, Membrane, Williams-Otto, Distillation, CSTR), bilevel BO matches or surpasses black-box BO's final performance within the initial sampling phase alone—effectively from the first BO iteration. Even in the hardest case (Evaporator), crossover occurs at 47 evaluations (19% of the total budget). This has direct practical implications: when each black-box evaluation costs hours of compute time, bilevel BO can deliver black-box BO-quality solutions using 80–100% fewer expensive evaluations.

Connection to decomposition in global optimization. As discussed in Section 2, the bilevel reformulation belongs to a broader family of decomposition strategies—block coordinate descent [32], trust-region filter methods [4, 13], and data-driven bilevel solvers like DOMINO [8]. Our results provide empirical support for a principle from bilevel optimization theory [33]: solving the inner problem exactly improves outer-level convergence. In the surrogate-based setting, the inner global optimizer eliminates noise from outer-level observations, giving the GP a cleaner signal. This suggests that bilevel decomposition may be beneficial whenever part of the optimization landscape admits an efficient exact solver, even beyond the grey-box problems studied here.

Complementarity with existing grey-box BO. As detailed in Section 2, COBALT [6] and BOCF [5] address settings our method does not: non-separable composite functions and risk-averse decisions via uncertainty propagation. Our method trades their generality for simplicity and exactness when Assumption 1 holds. A direct comparison on the same benchmark suite is planned for future work.

Baseline constraint handling. The black-box BO baseline uses a fixed penalty (10^6) for constraint violations. More sophisticated constrained BO methods—feasibility-weighted expected improvement [26], augmented Lagrangian approaches [34]—would likely narrow the gap on tightly constrained problems

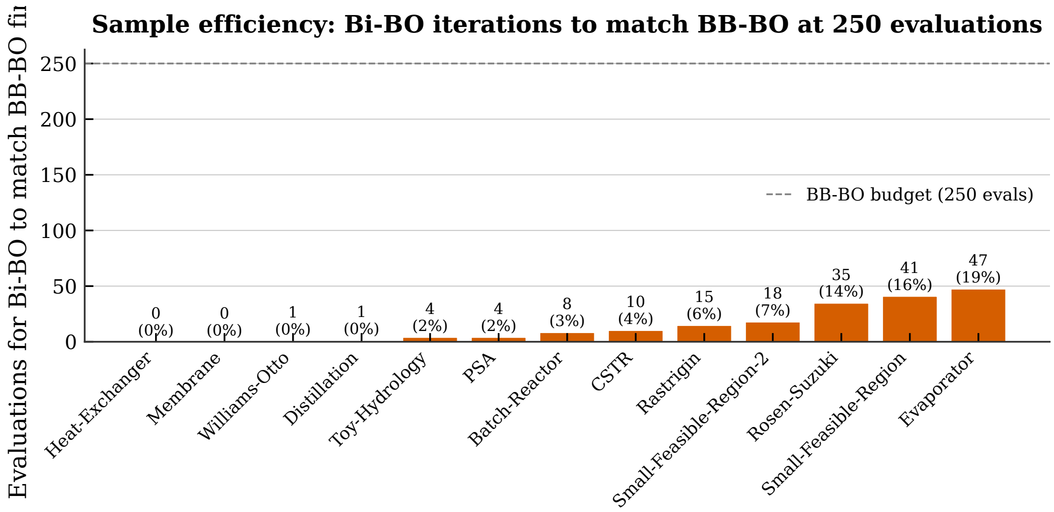

Figure 10: Sample efficiency crossover: number of evaluations for bilevel BO to match black-box BO's final regret (at 250 evaluations). Percentages show the fraction of the total budget required. On 10 of 13 problems, bilevel BO matches black-box BO's final performance within 10% of the evaluation budget.

such as Heat-Exchanger and Distillation, where the penalty method is least effective. However, the dimensionality reduction advantage of bilevel BO would persist regardless of constraint handling, as the surrogate dimension reduction from $n_{\mathrm{WB}} + n_{\mathrm{BB}}$ to n_{BB} is independent of the acquisition function. We intentionally use vanilla methods for both BO and NLP to isolate the effect of the bilevel structure rather than specific algorithmic choices.

Limitations and open problems. We identify five limitations, each pointing to an open problem. (1) The inner optimizer must converge to the global optimum (Assumption 1(ii)). For non-convex inner problems, DE may converge to local optima—as observed in Section 6, this explains both the modest 14× improvement on Rosen-Suzuki and the sub-100% feasibility on Batch-Reactor and Evaporator (Figure 8). Replacing DE with a deterministic global solver (e.g., BARON) would provide optimality certificates for the inner problem but at higher computational cost. (2) Expensive inner optimizations can dominate wall time, as Williams-Otto demonstrates (111s vs. 26s). Warm-starting the inner solver from previous solutions or using reduced-order inner models could mitigate this. (3) Pure black-box constraints—those depending directly on f^{BB} outputs without passing through the white-box model—are not handled; extending the outer loop with constrained BO methods [26, 27] would address this. (4) All black-box functions in our suite are closed-form surrogates of what would be expensive computations; validation on a real DFT or molecular dynamics black box would strengthen the practical case. (5) Formal convergence guarantees for the outer BO loop—which follow from standard results since it is BO over a compact domain with a GP surrogate—are not analyzed here; characterizing how the bilevel structure affects dimension-dependent regret bounds is future work.

Broader implications. Expensive black-box optimization problems frequently contain exploitable structure—known physics, analytical submodels, or differentiable components—that monolithic surrogates ignore. The bilevel decomposition is one instance of a general principle: *solve what you can solve ex-*

actly, and surrogate only what you must. As surrogate-based methods scale to higher-dimensional search spaces, structural decomposition becomes not merely helpful but necessary to overcome the curse of dimensionality. The 13-problem benchmark suite provides a testbed for developing and comparing methods that exploit this structure.

8 Conclusion

When grey-box optimization problems exhibit variable separability—the expensive black-box depends only on one variable group while the remaining variables enter known, differentiable equations—solving the known subproblem exactly and surrogating only the black-box component yields 14×–277,000× lower regret than monolithic surrogate-based optimization. The advantage grows with dimension, is robust to hyperparameters, and requires no algorithmic novelty beyond the decomposition itself. A suite of 13 benchmark problems with verified global optima provides a standardized testbed for future methods that exploit this structure.

The opening observation of this paper—that monolithic surrogates waste samples learning what is already known—is nearly self-evident once the separable structure is recognized. Yet no prior method systematically exploits this structure for dimensionality reduction with exact constraint satisfaction (in principle; subject to inner-solver global optimality in practice). The empirical evidence across 7,410 independent optimization runs confirms that the payoff is large and consistent. As expensive black-box optimization scales to higher dimensions, identifying and exploiting known substructure will be essential; the bilevel decomposition studied here is one principled way to do so.

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- **Conflict of interest.** The authors declare that they have no conflict of interest.

- **Data availability.** All data generated during this study are reproducible from the code described below. No external datasets were used.

- **Code availability.** The Python code for the benchmark suite, solvers, and experiment runner is available at .

- **Author contributions.** Joshua E. Hammond: conceptualization, methodology, software, investigation, writing—original draft. Tyler A. Soderstrom: methodology, investigation, writing—review & editing. Brian A. Korgel: supervision, writing—review & editing. Michael Baldea: conceptualization, supervision, funding acquisition, writing—review & editing.
- **Use of generative AI.** Generative AI tools (Claude Opus 4.6, Anthropic) were used to assist with code development, data analysis, and manuscript drafting. All AI-generated content was reviewed, verified, and edited by the authors, who take full responsibility for the accuracy and integrity of the work. No AI tool was used to generate scientific claims, interpret results, or design experiments without human oversight.

Appendix A Test Problem Definitions

Full mathematical formulations for all 13 benchmark problems. Each problem specifies the white-box and black-box models, constraints, variable bounds, and verified global optimum.

A.1 Small Feasible Region Problem 1

Gardner et al. [26] and Ariafar et al. [31] propose a constrained optimization test problem with a small feasible region that can be formulated as follows:

$$\begin{aligned} \min_x \; & \sin x_1 + y_1, \\ \text{s.t.} \; & g_1(x) := \sin(x_1)\sin(y_1) + 0.95 \le 0, \\ & y_1 = f^{\mathrm{BB}}(x) := x_2, \\ & 0 \le x_i \le 6, \quad \forall i \in \{1,2\} \end{aligned} \tag{6}$$

The global minimum is $J^\star = 0.2532$ and is located at $x_1^{\mathrm{WB}\star} = \frac{3\pi}{2}$, $x^{\mathrm{BB}\star} = x_2 = 1.2532$.

A.2 Modified Small Feasible Region Problem 2

Gardner et al. [26] and Ariafar et al. [31] propose a constrained optimization test problem with a small feasible region. We add an additional black-box function $y_1 = f^{\mathrm{BB}}(x_2)$ that forms a grey-box problem as follows:

$$\begin{aligned} \min_x \; & \cos(2x_1)\cos(y_1) + \sin(x_1), \\ \text{s.t.} \; & g_1(x) := \cos(x_1)\cos(y_1) - \sin(x_1)\sin(y_1) - 0.5 \le 0, \\ & y_1 = f^{\mathrm{BB}}(x) := \frac{(x_2 - 5.5)}{2}\exp\left(\frac{x_2}{2}\right), \\ & 0 \le x_i \le 6, \quad \forall i \in \{1,2\} \end{aligned} \tag{7}$$

The global minimum $J^\star = -2.0$ is located at $x^{\mathrm{BB}\star} = 5.5, x^{\mathrm{WB}\star} = 3\pi/2$.

A.3 Rastrigin

We use a modified formulation of the Rastrigin function [35] that can be formulated as a multi-scale grey-box problem as follows:

$$\begin{aligned}
\min_{x,y} \quad & 30 + x_1^2 - 10\cos(2\pi x_1) + x_2^2 - 10\cos(2\pi x_2) + y_1, \\
\text{s.t.} \quad & y_1 = f^{\text{BB}}(x) := x_3^2 - 10\cos(2\pi x_3), \\
& -5.12 \le x_i \le 5.12, \qquad \forall i \in \{1,2,3\}, \\
& x = [x^{\text{WB}}, x^{\text{BB}}], \qquad x^{\text{WB}} = [x_1, x_2], \qquad x^{\text{BB}} = [x_3].
\end{aligned} \tag{8}$$

The global minimum is equal to 0 with $x^{\text{WB}\star} = [0,0], x^{\text{BB}\star} = [0]^\top$.

A.4 Toy-Hydrology

We use a modified formulation of the Toy Hydrology problem [34] that can be formulated as a multi-scale grey-box problem as follows:

$$\begin{aligned}
\min_{x,y} \quad & x_1 + x_2, \\
\text{s.t.} \quad & g_1(x,y) := 1.5 - x_1 - 2x_2 - 0.5\sin(-4\pi x_2 + y_1) \le 0, \\
& g_2(x,y) := x_1^2 + x_2^2 - 1.5 \le 0, \\
& y_1 = f^{\text{BB}}(x_1) := 2\pi x_1^2, \\
& 0 \le x_i \le 1, \qquad \forall i \in \{1,2\}, \\
& x = [x^{\text{WB}}, x^{\text{BB}}], \qquad x^{\text{BB}} = [x_1], \qquad x^{\text{WB}} = [x_2].
\end{aligned} \tag{9}$$

The global minimum is 0.5998 with $x^{\text{WB}\star} = 0.4047, x^{\text{BB}\star} = 0.1951$.

A.5 Rosen-Suzuki

We use a modified formulation of the Rosen-Suzuki function [36] that can be formulated as a multi-scale grey-box problem as follows:

$$\begin{aligned}
\min_{x,y} \quad & x_1^2 + x_2^2 + x_4^2 - 5x_1 - 5x_2 + y_1, \\
\text{s.t.} \quad & g_1(x,y) := -(8 - x_1^2 - x_2^2 - x_3^2 - x_4^2 - x_1 + x_2 - x_3 + x_4) \le 0, \\
& g_2(x,y) := -(10 - x_1^2 - 2x_2^2 - y_2 + x_1 + x_4) \le 0, \\
& g_3(x,y) := -(5 - 2x_1^2 - x_2^2 - x_3^2 - 2x_1 + x_2 + x_4) \le 0, \\
& y_1 = f_1^{\text{BB}}(z) := 2x_3^2 - 21x_3 + 7x_4, \\
& y_2 = f_2^{\text{BB}}(z) := x_3^2 + 2x_4^2, \\
& -2 \le x_i \le 2, \qquad \forall i \in \{1,\dots,4\} \\
& x = [x^{\text{WB}}, x^{\text{BB}}], \qquad x^{\text{WB}} = [x_1, x_2], \qquad x^{\text{BB}} = [x_3, x_4].
\end{aligned} \tag{10}$$

The global minimum is -44 with $x^{\text{WB}\star} = [0,1]^\top, x^{\text{BB}\star} = [2,-1]^\top$.

A.6 Catalytic Reactor Design

We use a catalytic reactor design problem that combines a white-box CSTR process model with black-box catalyst properties akin to those derived from

DFT calculations. The objective is to *maximize* the yield of intermediate product B in consecutive reactions $A \to B \to C$. The problem can be formulated as a multi-scale grey-box problem as follows:

$$
\begin{aligned}
\max_{T,P,\tau,E_b,\Delta E_a} \quad & Y_B \qquad\qquad (11)\\
\text{s.t.} \quad & \mathbf{f}^{\mathrm{WB}} = \begin{bmatrix} Y_B - \frac{k_1\tau}{(1+k_1\tau)(1+k_2\tau)} \\ k_1\left(T,P,E_b,\Delta E_a\right) - A_1 v\left(E_b\right)\exp\left(-\frac{E_{a,1}}{RT}\right)\left(\frac{P}{P_{\mathrm{ref}}}\right)^{\beta_1}\frac{1}{(1+K_{\mathrm{ads}}P)^2} \\ k_2\left(T,P,E_b,\Delta E_a\right) - A_2\exp\left(-\frac{E_{a,2}}{RT}\right)\left(\frac{P}{P_{\mathrm{ref}}}\right)^{\beta_2}\frac{1}{(1+K_{\mathrm{ads}}P)^2} \\ K_{\mathrm{ads}}\left(E_b\right) - K_0\exp\left(\kappa\left(E_b - E_b^*\right)\right) \end{bmatrix}, \\
& \mathbf{f}^{\mathrm{BB}} = \begin{bmatrix} v\left(E_b\right) - \left[\exp\left(-\frac{1}{2}\left(\frac{E_b - E_b^*}{\sigma_E}\right)^2\right)\right] \\ E_{a,1} - \left[E_{a,1}^{(0)} - \alpha_1\Delta E_a\right] \\ E_{a,2} - \left[E_{a,2}^{(0)} + \alpha_2\Delta E_a\right] \end{bmatrix}, \\
& g_1(x,y) := Y_C - 0.25 \le 0, \\
& g_2(x,y) := \frac{P}{P_{\max}} - \frac{T - T_{\min}}{T_{\max} - T_{\min}} - 0.3 \le 0,
\end{aligned}
$$

and the conversion and byproduct yield are:

$$
X_A = \frac{k_1\tau}{1+k_1\tau}, \qquad\qquad Y_C = X_A - Y_B.
$$

The black-box function f^{BB} maps catalyst properties ΔE_a, E_b to kinetic parameters via a volcano-type activity model:

The global maximum yield given the parameters in Table 4 is 92.32% with $x^{\mathrm{WB}\star} = [647.3, 20.0, 5.0]^\top$, $x^{\mathrm{BB}\star} = [-1.047, 0.200]^\top$.

A.7 Heat Exchanger with Novel Working Fluid

We consider the design of a heat exchanger system where the working fluid's thermodynamic properties (heat capacity c_p, viscosity μ, and thermal conductivity k) depend on two molecular descriptors (m_1, m_2) in a black-box function. White-box equations govern the heat exchanger design and operation (heat transfer area A, mass flow rate $\dot{m}$, and log-mean temperature difference ΔT_{lm}) with the goal of minimizing the total cost (capital + operating) while satisfying minimum heat duty [37]. Capital cost scales with area according to the six-tenths rule [38] while operating cost is a function of pumping power which depends on fluid viscosity and mass flow rate according to the relation provided by [39]. We use a simplified form for the overall heat transfer coefficient U based the resistance model from Ravagnani and Caballero [40].

Table 4: Parameters for the catalytic reactor design problem.

Symbol	**Description**	**Range / Value**	**Units**
Decision / design variables			
T	Temperature	[450, 700]	K
P	Pressure	[1, 20]	bar
τ	Residence time	[0.1, 5.0]	s
E_b	Binding energy	[-2.0, 0.0]	eV
ΔE_a	Activation-energy shift	[-0.2, 0.2]	-
A_1	Pre-exponential (step 1)	1.0×10^7	s^{-1}
A_2	Pre-exponential (step 2)	3.0×10^6	s^{-1}
$E_{a1}\,(\Delta E_a)$	Activation energy	$80,000 - 20,000\Delta E_a$	J/mol
$E_{a2}\,(\Delta E_a)$	Activation energy	$95,000 + 5,000\Delta E_a$	J/mol
P_{ref}	Reference pressure	8.0	bar
$E_b^\star$	Reference binding energy	-1.0	eV
σ_E	Width parameter	0.40	eV
K_0	Adsorption rate pre-exponential factor	0.06	1/bar
κ	Adsorption rate exponential term	3.0	1/eV
R	Gas constant	8.314	J/(mol K)
k_B	Boltzmann constant	8.617×10^{-5}	eV/K
α_1		20,000	-
α_2		5,000	-
β_1	Exponent for pressure dependence	1.0	-
β_2	Exponent for pressure dependence	0.35	-

The problem can be formulated as:

$$
\begin{aligned}
\min_{A,\dot{m},\Delta T_{lm},m_1,m_2} \quad & C_{\text{capital}} + C_{\text{operating}} \\
\text{s.t.} \quad & C_{\text{capital}} = 1000 \cdot x_1^{\text{WB}\,0.6}, \\
& C_{\text{operating}} = 500 \cdot \frac{x_2^{\text{WB}\,3} \cdot y_2}{x_1^{\text{WB}}}, \\
& U = \frac{y_3}{0.01 + 0.001/y_1}, \\
& g_1(x,y) := 100 - U \cdot x_1^{\text{WB}} \cdot x_3^{\text{WB}} \leq 0, \\
& g_2(x,y) := 4000 - \frac{4x_2^{\text{WB}}}{\pi \cdot 0.05 \cdot y_2} \leq 0, \\
& y_1 = f_1^{\text{BB}}(x_1^{\text{BB}}, x_2^{\text{BB}}) := 2.0 + 0.5 \sin\left(\frac{\pi x_1^{\text{BB}}}{50}\right) + 0.3 x_2^{\text{BB}}, \\
& y_2 = f_2^{\text{BB}}(x_1^{\text{BB}}, x_2^{\text{BB}}) := 0.001 \cdot \exp\left(\frac{x_1^{\text{BB}} - 100}{50}\right) \cdot (1 + 0.5 x_2^{\text{BB}}), \\
& y_3 = f_3^{\text{BB}}(x_1^{\text{BB}}, x_2^{\text{BB}}) := 0.1 + 0.05\left(1 - \left(\frac{x_1^{\text{BB}} - 125}{75}\right)^2\right) + 0.02 x_2^{\text{BB}}, \\
& A \in [1, 50], \quad \dot{m} \in [0.1, 5], \quad \Delta T_{lm} \in [5, 50], \\
& m_1 \in [50, 200], \quad m_2 \in [0, 1], \\
& x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [A, \dot{m}, \Delta T_{lm}], \quad x^{\text{BB}} = [m_1, m_2],
\end{aligned} \tag{12}
$$

Table 5: Variables, physical meaning, bounds, and units (Heat exchanger)

Variable	Physical meaning	Lower	Upper	Units
x_1^{WB}	Heat transfer area (A)	1	50	m^2
x_2^{WB}	Mass flow rate ($\dot{m}$)	0.1	5	kg/s
x_3^{WB}	Log-mean temperature difference (ΔT_{lm})	5	50	K
x_1^{BB}	Molecular descriptor (m_1)	50	200	(unitless)
x_2^{BB}	Molecular descriptor (m_2)	0	1	(unitless)
y_1	Heat capacity (c_p)	N/A	N/A	kJ/kg·K
y_2	Viscosity (μ)	N/A	N/A	Pa·s
y_3	Thermal conductivity (k)	N/A	N/A	W/m·K

where y_1 is the heat capacity [kJ/kg·K], y_2 is the viscosity [Pa·s], and y_3 is the thermal conductivity [W/m·K]. The constraint g_1 ensures the heat duty $Q = U \cdot A \cdot \Delta T_{lm} \geq 100$ kW is satisfied, and g_2 enforces turbulent flow with Reynolds number $\text{Re} \geq 4000$. The global minimum cost given the parameters in Table 5 is 1000.0 with $x^{\text{WB}\star} = [1.0, 0.1, 19.3]^\top$, $x^{\text{BB}\star} = [50.0, 0.0]^\top$.

A.8 Pressure Swing Adsorption with Novel Adsorbent

We consider the design of a pressure swing adsorption (PSA) cycle for gas separation, a problem that naturally exhibits grey-box structure [3, 41, 42]. The adsorbent properties—maximum loading $q_{\max}$, adsorption equilibrium constant for the target component K_A, and equilibrium constant for the competing species K_B—are computed from molecular simulations [43, 44] that serve as black-box functions of adsorbent structural parameters (pore size σ and interaction energy ϵ) [45].

The white-box model captures the PSA cycle performance through Langmuir isotherm-based working capacity calculations [46, 47]. The working capacity Δq represents the difference in loading between high-pressure adsorption and low-pressure desorption steps [48, 42]. Selectivity $\alpha = K_A/K_B$ determines separation performance [49], while productivity is defined as working capacity divided by adsorption time [50]. The objective minimizes the trade-off between maximizing productivity and minimizing compression costs, where isothermal compression work scales with $\ln(P_H/P_L)$ [51].

The problem can be formulated as:

$$
\begin{aligned}
\min_{P_H, P_L, t_{\mathrm{ads}}, \sigma, \epsilon} \quad & -\mathrm{Prod} + 0.1 \cdot \ln\left(\frac{P_H}{P_L}\right) \\
\text{s.t.} \quad & \alpha = \frac{y_2}{y_3}, \\
& \Delta q = y_1 \cdot \left(\frac{y_2 \cdot P_H}{1 + y_2 \cdot P_H} - \frac{y_2 \cdot P_L}{1 + y_2 \cdot P_L}\right), \\
& \mathrm{Prod} = \frac{\Delta q}{t_{\mathrm{ads}}}, \\
& g_1(x, y) := 3 - \alpha \leq 0, \\
& g_2(x, y) := 0.95 - \frac{\alpha \cdot P_H}{1 + \alpha \cdot P_H} \leq 0, \\
& y_1 = f_1^{\mathrm{BB}}(\sigma, \epsilon) := 5.0 \cdot \exp\left(-\frac{(\sigma - 6)^2}{4}\right) \cdot (1 + 0.1\epsilon), \\
& y_2 = f_2^{\mathrm{BB}}(\sigma, \epsilon) := 0.5 \cdot \exp\left(\frac{\epsilon - 10}{5}\right) \cdot \left(1 - 0.05(\sigma - 5)^2\right), \\
& y_3 = f_3^{\mathrm{BB}}(\sigma, \epsilon) := 0.1 \cdot \exp\left(\frac{\epsilon - 15}{8}\right) \cdot \left(1 + 0.03(\sigma - 7)^2\right), \\
& P_H \in [2, 10], \quad P_L \in [0.1, 1], \quad t_{\mathrm{ads}} \in [10, 120], \\
& \sigma \in [3, 10], \quad \epsilon \in [5, 25], \\
& x = [x^{\mathrm{WB}}, x^{\mathrm{BB}}], \quad x^{\mathrm{WB}} = [P_H, P_L, t_{\mathrm{ads}}], \quad x^{\mathrm{BB}} = [\sigma, \epsilon],
\end{aligned}
\tag{13}
$$

The black-box functions f^{BB} model the adsorbent properties as functions of the Lennard-Jones parameters: pore size σ [Å] and interaction energy ϵ [kJ/mol] [45, 44]. These relationships capture how molecular-scale adsorbent design affects macroscopic separation performance [43].

The constraint g_1 ensures minimum selectivity $\alpha \geq 3$ for effective separation [49, 48], and g_2 enforces minimum product purity based on the Langmuir isotherm equilibrium [46, 42]. The global minimum given the parameters in Table 6 is -0.6433 with $x^{\mathrm{WB}\star} = [3.97, 0.1, 10.0]^\top$, $x^{\mathrm{BB}\star} = [6.04, 19.55]^\top$.

A.9 Batch Reactor with Catalyst Optimization

We consider the optimization of a batch reactor for consecutive reactions $A \to B \to C$, where the objective is to maximize the yield of the intermediate product B [52]. This classic selectivity problem [53] gains a multi-scale

Table 6: Variables, physical meaning, bounds, and units (Pressure Swing Adsorption)

Symbol	Description	Range / Value	Units
White-box decision variables			
P_H	High (adsorption) pressure	[2, 10]	bar
P_L	Low (desorption) pressure	[0.1, 1]	bar
t_{ads}	Adsorption step time	[10, 120]	s
Black-box decision variables			
σ	Adsorbent pore size	[3, 10]	Å
ϵ	Lennard-Jones interaction energy	[5, 25]	kJ/mol
Black-box outputs			
$y_1 = q_{\mathrm{max}}$	Maximum loading capacity	—	mol/kg
$y_2 = K_A$	Adsorption equilibrium constant (component A)	—	1/bar
$y_3 = K_B$	Adsorption equilibrium constant (component B)	—	1/bar

dimension when catalyst properties must be co-optimized with operating conditions.

The white-box model consists of standard Arrhenius kinetics [54] governing the batch reactor mass balances, which admit analytical solutions for first-order consecutive reactions. Temperature T, batch time t_f, and initial concentration $C_{A,0}$ are the white-box decision variables.

The black-box model implements a Sabatier volcano relationship [55, 56, 57] that captures how catalyst activity depends on binding energy E_b. The volcano principle states that optimal catalysts bind reactants neither too strongly (limiting desorption) nor too weakly (limiting adsorption), with peak activity at an intermediate binding energy. The catalyst particle size d provides an additional degree of freedom that affects the pre-exponential factor through available surface area.

The problem can be formulated as:

$$
\begin{aligned}
\min_{T,t_f,C_{A,0},E_b,d} \quad & -C_B + 0.001 \cdot T \cdot t_f && (14)\\
\text{s.t.} \quad & k_1' = y_1 \cdot \exp\left(\frac{-2000}{T}\right), \\
& k_2' = y_2 \cdot \exp\left(\frac{-2400}{T}\right), \\
& C_A = C_{A,0} \cdot \exp(-k_1' \cdot t_f), \\
& C_B = C_{A,0} \cdot \frac{k_1'}{k_2' - k_1'} \cdot \left(\exp(-k_1' \cdot t_f) - \exp(-k_2' \cdot t_f)\right), \\
& g_1(x,y) := 0.8 - \left(1 - \frac{C_A}{C_{A,0}}\right) \leq 0, \\
& g_2(x,y) := 0.7 - \frac{C_B}{C_{A,0} - C_A} \leq 0, \\
& y_1 = f_1^{\mathrm{BB}}(E_b, d) := 10 \cdot d \cdot \exp\left(-\frac{(E_b+1)^2}{0.25}\right), \\
& y_2 = f_2^{\mathrm{BB}}(E_b, d) := 0.5 \cdot d \cdot \exp\left(-\frac{(E_b+0.5)^2}{0.5}\right), \\
& y_3 = f_3^{\mathrm{BB}}(E_b, d) := \exp\left(-2 \cdot E_b - 1\right), \quad \text{(not used in white-box equations)} \\
& T \in [300, 500], \quad t_f \in [0.5, 10], \quad C_{A,0} \in [0.5, 5], \\
& E_b \in [-2, 0], \quad d \in [0.5, 2], \\
& x = [x^{\mathrm{WB}}, x^{\mathrm{BB}}], \quad x^{\mathrm{WB}} = [T, t_f, C_{A,0}], \quad x^{\mathrm{BB}} = [E_b, d],
\end{aligned}
$$

The Gaussian functions in f_1^{BB} and f_2^{BB} encode the volcano-shaped activity curves: k_1 peaks at $E_b = -1$ eV while k_2 peaks at $E_b = -0.5$ eV, creating a selectivity landscape where catalyst design can favor the desired reaction over the consecutive side reaction. The output y_3 does not appear in any white-box equation or constraint; it is included intentionally to test robustness to irrelevant black-box outputs, since in practice molecular simulations often return quantities not all of which enter the process model.

The constraint g_1 ensures minimum conversion $X_A \geq 0.8$, and g_2 enforces selectivity $S_B = C_B/(C_{A,0} - C_A) \geq 0.7$, requiring that at least 70% of converted reactant forms the desired intermediate. The global minimum given the parameters in Table 7 is -1.7488 with $x^{\mathrm{WB}\star} = [500.0, 4.394, 5.0]^\top$, $x^{\mathrm{BB}\star} = [-1.01, 2.0]^\top$.

A.10 Extractive Distillation with Novel Solvent

We consider the integrated design of an extractive distillation column and entrainer selection, a canonical example of simultaneous solvent and process optimization [58, 59]. The white-box model employs classical shortcut methods: the Fenske equation [60] determines minimum stages $N_{\min}$ from the relative volatility and separation specifications, the Underwood equations [61, 62] yield minimum reflux $R_{\min}$, and operating constraints follow from the Gilliland correlation [63, 64].

The black-box model represents COSMO-RS or other molecular thermodynamic predictions that map solvent molecular descriptors—here parameterized by Hansen solubility parameters for hydrogen bonding δ_H and polar

Table 7: Variables, physical meaning, bounds, and units (Batch Reactor)

Symbol	**Description**	**Range / Value**	**Units**
White-box decision variables			
T	Reactor temperature	[300, 500]	K
t_f	Batch (final) time	[0.5, 10]	h
$C_{A,0}$	Initial concentration of A	[0.5, 5]	mol/L
Black-box decision variables			
E_b	Catalyst binding energy	$[-2, 0]$	eV
d	Catalyst particle size factor	[0.5, 2]	—
Black-box outputs			
$y_1 = k_1^0$	Pre-exponential factor for $A \to B$	—	1/s
$y_2 = k_2^0$	Pre-exponential factor for $B \to C$	—	1/s
$y_3 = K_{\text{eq}}$	Adsorption equilibrium constant	—	—

interactions δ_P—to the modified relative volatility α_{12}^{mod} in the presence of the entrainer [65]. The entrainer also affects operating costs through its density ρ (determining column sizing) and viscosity μ (affecting tray hydraulics and heat transfer).

The problem can be formulated as:

$$\begin{aligned}
\min_{R,N,S/F,\delta_H,\delta_P} \quad & 1000 \cdot N + 500 \cdot R + 200 \cdot (S/F) \cdot y_3 \qquad (15)\\
\text{s.t.} \quad & R_{\min} = \frac{1}{y_1 - 1} \cdot \left(\frac{x_D}{x_F} - y_1 \cdot \frac{1 - x_D}{1 - x_F} \right), \\
& N_{\min} = \frac{\ln\left(\frac{x_D(1-x_B)}{x_B(1-x_D)}\right)}{\ln(y_1)}, \\
& g_1(x, y) := 1.2 \cdot R_{\min} - R \leq 0, \\
& g_2(x, y) := 1.5 \cdot N_{\min} - N \leq 0, \\
& g_3(x, y) := (S/F) \cdot y_2 - 5000 \leq 0, \\
& y_1 = f_1^{\text{BB}}(\delta_H, \delta_P) := 2.5 + 1.0 \cdot \sin\left(\frac{\pi(\delta_H - 10)}{20}\right) \cdot \cos\left(\frac{\pi(\delta_P - 10)}{15}\right), \\
& y_2 = f_2^{\text{BB}}(\delta_H, \delta_P) := 800 + 50 \cdot \delta_H - 20 \cdot \delta_P, \\
& y_3 = f_3^{\text{BB}}(\delta_H, \delta_P) := 0.5 + 0.1 \cdot \delta_H + 0.05 \cdot \delta_P^2/100, \\
& R \in [1, 10], \quad N \in [5, 50], \quad S/F \in [0.5, 5], \\
& \delta_H \in [5, 25], \quad \delta_P \in [5, 20], \\
& x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [R, N, S/F], \quad x^{\text{BB}} = [\delta_H, \delta_P],
\end{aligned}$$

where $x_D = 0.99$ (distillate purity), $x_F = 0.5$ (feed composition), and $x_B = 0.01$ (bottoms impurity).

The constraint g_1 ensures the reflux ratio exceeds the minimum by at least 20% for operational flexibility, g_2 requires sufficient stages above the Fenske minimum, and g_3 limits solvent inventory based on density to control capital costs. The global minimum cost for the parameters in Table 8 is 11,758 with $x^{\text{WB}\star} = [1.0, 11.0, 0.5]^\top$, $x^{\text{BB}\star} = [19.8, 9.99]^\top$.

Table 8: Variables, physical meaning, bounds, and units (Extractive Distillation)

Symbol	**Description**	**Range / Value**	**Units**
White-box decision variables			
R	Reflux ratio	[1, 10]	—
N	Number of theoretical stages	[5, 50]	—
S/F	Solvent-to-feed ratio	[0.5, 5]	—
Black-box decision variables			
δ_H	Hansen hydrogen-bonding parameter	[5, 25]	$\mathrm{MPa}^{0.5}$
δ_P	Hansen polar parameter	[5, 20]	$\mathrm{MPa}^{0.5}$
Black-box outputs			
$y_1 = \alpha_{12}^{\mathrm{mod}}$	Modified relative volatility	—	—
$y_2 = \rho$	Solvent density	—	$\mathrm{kg/m}^3$
$y_3 = \mu$	Solvent viscosity	—	cP
Fixed parameters			
x_D	Distillate purity specification	0.99	mol/mol
x_F	Feed composition	0.50	mol/mol
x_B	Bottoms impurity specification	0.01	mol/mol

A.11 Multi-Effect Evaporator with Phase-Change Material

We consider a prototype test problem for the design of a multi-effect evaporator (MEE) system integrated with a novel phase-change material (PCM) as the heat transfer medium. Multi-effect evaporator synthesis follows established energy balance formulations [66], where steam economy improves with additional effects but with diminishing returns due to tighter temperature driving forces. The efficiency degradation across effects is captured by the factor 0.9^{n-1}, consistent with typical industrial values of 0.7–0.9 pounds of evaporation per pound of steam per effect [67].

The white-box model includes energy balances relating the heat transfer fluid (HTF) flow rate and properties to evaporation capacity [68]. The latent heat of vaporization for water (2260 kJ/kg) determines the evaporation rate from available thermal energy. The number of effects n, HTF mass flow rate $\dot{m}_{\mathrm{HTF}}$, and temperature drop per effect $\Delta T_{\mathrm{effect}}$ are the white-box decision variables.

The black-box model represents molecular simulation or group-contribution predictions of PCM thermophysical properties [69, 70] as functions of melting temperature T_m and a molecular structure parameter λ that controls the length of alkyl chains (affecting both latent heat and heat capacity). The key outputs are latent heat of fusion ΔH_{fus} (typical range 150–300 kJ/kg for organic PCMs), liquid heat capacity c_p^{liq} (typically 2–3 kJ/(kg·K)), and the actual melting temperature T_{melt}. The functional forms in f^{BB} are intended to capture qualitative structure-property relationships rather than precise physical models.

The problem can be formulated as:

$$
\begin{aligned}
\min_{n,\dot{m}_{\mathrm{HTF}},\Delta T_{\mathrm{effect}},T_m,\lambda} \quad & -\dot{m}_{\mathrm{evap}} + 0.1\cdot n^2 + 0.01\cdot \dot{m}_{\mathrm{HTF}}^2 \\
\text{s.t.} \quad & Q_{\mathrm{HTF}} = \dot{m}_{\mathrm{HTF}}\cdot(y_1 + y_2\cdot 10), \\
& \dot{m}_{\mathrm{evap}} = \frac{Q_{\mathrm{HTF}}\cdot n\cdot 0.9^{n-1}}{2260}, \\
& g_1(x,y) := 100 + n\cdot \Delta T_{\mathrm{effect}} - y_3 \le 0, \\
& g_2(x,y) := 5 - \dot{m}_{\mathrm{evap}} \le 0, \\
& y_1 = f_1^{\mathrm{BB}}(T_m,\lambda) := 150\cdot\lambda\cdot\left(1 + 0.2\sin\left(\frac{\pi T_m}{100}\right)\right), \\
& y_2 = f_2^{\mathrm{BB}}(T_m,\lambda) := 2.0 + 0.5\cdot\lambda - 0.002\cdot T_m, \\
& y_3 = f_3^{\mathrm{BB}}(T_m,\lambda) := T_m + 10\cdot\sin\left(\frac{\pi\lambda}{2}\right), \\
& n\in[2,6] \text{ (treated as continuous; integer rounding would apply in practice)}, \\
& \dot{m}_{\mathrm{HTF}}\in[1,20], \quad \Delta T_{\mathrm{effect}}\in[5,20], \\
& T_m\in[50,150], \quad \lambda\in[0.5,2], \\
& x = [x^{\mathrm{WB}}, x^{\mathrm{BB}}], \quad x^{\mathrm{WB}} = [n, \dot{m}_{\mathrm{HTF}}, \Delta T_{\mathrm{effect}}], \quad x^{\mathrm{BB}} = [T_m,\lambda],
\end{aligned}
\tag{16}
$$

Table 9: Variables, physical meaning, bounds, and units (Multi-Effect Evaporator)

Symbol	**Description**	**Range / Value**	**Units**
White-box decision variables			
n	Number of evaporator effects	[2, 6]	—
$\dot{m}_{\mathrm{HTF}}$	Heat transfer fluid mass flow rate	[1, 20]	kg/s
$\Delta T_{\mathrm{effect}}$	Temperature drop per effect	[5, 20]	°C
Black-box decision variables			
T_m	Target melting temperature	[50, 150]	°C
λ	Molecular structure parameter	[0.5, 2]	—
Black-box outputs			
$y_1 = \Delta H_{\mathrm{fus}}$	Latent heat of fusion	—	kJ/kg
$y_2 = c_p^{\mathrm{liq}}$	Liquid phase heat capacity	—	kJ/(kg·K)
$y_3 = T_{\mathrm{melt}}$	Actual melting temperature	—	°C
Fixed parameters			
$\Delta H_{\mathrm{vap,water}}$	Latent heat of water vaporization	2260	kJ/kg
η_{effect}	Per-effect efficiency factor	0.9	—

The constraint g_1 ensures temperature feasibility: the PCM melting temperature must exceed the cumulative temperature drop across all effects plus a 100°C baseline. The constraint g_2 enforces a minimum evaporation rate of 5 kg/s to meet production requirements. The global minimum for parameters in Table 9 is -1.9587 with $x^{\mathrm{WB}\star} = [4.09, 19.07, 5.0]^\top$, $x^{\mathrm{BB}\star} = [120.47, 2.0]^\top$.

A.12 Membrane Separation with Designed Polymer

We consider the integrated design of a membrane gas separation process and the polymer membrane material [71]. This problem exemplifies the permeability-selectivity trade-off characterized by the Robeson upper bound [72, 73], which establishes that polymeric membranes exhibit a fundamental inverse relationship between gas permeability and selectivity explained by free volume theory [74].

The white-box model follows the solution-diffusion transport mechanism [75], where the permeate flux J_A is proportional to permeability, pressure driving force, and inversely proportional to membrane thickness δ. The membrane area A_m, thickness δ, and transmembrane pressure difference Δp are the white-box decision variables, with the objective of maximizing product recovery while minimizing capital (area) and operating (compression) costs.

The black-box model represents structure-property relationships from molecular simulations or machine learning predictions [76] that map polymer descriptors—fractional free volume (FFV) and chain spacing d_{spacing}—to transport properties. The Robeson trade-off is encoded in the black-box functions: increasing FFV enhances permeability but reduces selectivity, while chain spacing affects both the upper bound position and mechanical integrity. The mechanical strength constraint reflects the requirement that the membrane must withstand the transmembrane pressure difference during module operation [77].

The problem can be formulated as:

$$
\begin{aligned}
\min_{A_m,\delta,\Delta p,\text{FFV},d_{\text{spacing}}} \quad & -F_A \cdot y_2 + 0.1 \cdot A_m + 10 \cdot \Delta p \qquad (17)\\
\text{s.t.} \quad & J_A = \frac{y_1 \cdot \Delta p}{\delta} \cdot 10^{-10},\\
& F_A = J_A \cdot A_m,\\
& g_1(x,y) := 10^{-6} - J_A \le 0,\\
& g_2(x,y) := \frac{\Delta p \cdot A_m}{y_3} - 100 \le 0,\\
& y_1 = f_1^{\text{BB}}(\text{FFV}, d_{\text{spacing}}) := 1000 \cdot \exp\left(\frac{\text{FFV} - 0.15}{0.05}\right) \cdot (1 + 0.1 \cdot d_{\text{spacing}}),\\
& y_2 = f_2^{\text{BB}}(\text{FFV}, d_{\text{spacing}}) := 50 \cdot \exp\left(-\frac{\text{FFV} - 0.15}{0.1}\right) \cdot \exp\left(-\frac{(d_{\text{spacing}} - 5)^2}{4}\right),\\
& y_3 = f_3^{\text{BB}}(\text{FFV}, d_{\text{spacing}}) := 100 \cdot (0.4 - \text{FFV}) \cdot (10 - d_{\text{spacing}}),\\
& A_m \in [10, 1000], \quad \delta \in [0.1, 10], \quad \Delta p \in [1, 20],\\
& \text{FFV} \in [0.1, 0.3], \quad d_{\text{spacing}} \in [3, 8],\\
& x = [x^{\text{WB}}, x^{\text{BB}}], \quad x^{\text{WB}} = [A_m, \delta, \Delta p], \quad x^{\text{BB}} = [\text{FFV}, d_{\text{spacing}}],
\end{aligned}
$$

Refer to the parameter definitions in Table 10.

The white-box flux equation $J_A = y_1 \cdot \Delta p/\delta$ follows directly from the solution-diffusion model [75], where permeate flux is proportional to permeability and pressure driving force, and inversely proportional to membrane thickness. The factor 10^{-10} converts from Barrer (the conventional permeability unit) to SI flux units.

The black-box functions are constructed to capture established structure-property relationships from polymer physics. The permeability function y_1

Table 10: Variables, physical meaning, bounds, and units (Membrane Separation)

Symbol	**Description**	**Range / Value**	**Units**
White-box decision variables			
A_m	Membrane area	[10, 1000]	m^2
δ	Membrane thickness	[0.1, 10]	μm
Δp	Transmembrane pressure difference	[1, 20]	bar
Black-box decision variables			
FFV	Fractional free volume	[0.1, 0.3]	—
d_{spacing}	Polymer chain spacing	[3, 8]	Å
Black-box outputs			
$y_1 = P_A$	Permeability of component A	—	Barrer
$y_2 = \alpha_{A/B}$	Selectivity (A over B)	—	—
$y_3 = \sigma$	Mechanical strength proxy	—	MPa

follows the exponential free volume dependence derived by Cohen and Turnbull [78] and extended by Fujita [79], who showed that diffusivity (and hence permeability) scales as $P \propto \exp(B \cdot \text{FFV})$ where B is a polymer-penetrant parameter. This exponential form arises because molecular transport occurs through transient gaps in the polymer matrix, with the probability of sufficiently large gaps increasing exponentially with free volume fraction [80]. The chain spacing term provides a secondary contribution reflecting the effect of interchain distance on diffusion pathways [81].

The selectivity function y_2 encodes the Robeson upper bound trade-off [72, 73]: as FFV increases, permeability rises but selectivity decreases because larger free volume elements become less size-discriminating. Freeman [74] provided a theoretical basis showing that the slope of the permeability-selectivity trade-off depends only on penetrant size ratios. The Gaussian dependence on d_{spacing} reflects an optimal chain spacing for size-sieving selectivity [77].

The mechanical strength proxy y_3 captures the physical principle that increased free volume and chain spacing reduce polymer density and chain entanglement, thereby weakening mechanical integrity [82]. While the linear functional form is a simplification, it correctly represents the inverse relationship between transport-enhancing microstructure and mechanical robustness that constrains practical membrane design [83].

The constraint g_1 ensures minimum flux $J_A \geq 10^{-6}$ mol/(m^2·s) for practical separation rates, and g_2 enforces mechanical stability by limiting the pressure-area product relative to membrane strength.

The global optimum for parameters in Table 10 occurs at $x^{\text{WB}*} = [A_m, \delta, \Delta p] = [10, 0.1, 1]$ and $x^{\text{BB}*} = [\text{FFV}, d_{\text{spacing}}] = [0.3, 5.13]$, yielding $J^* = 10.997$. The optimizer drives all white-box variables to their lower bounds—minimizing membrane area (capital cost) and pressure difference (operating cost) while maximizing flux through minimum thickness—and selects maximum FFV for permeability while tuning chain spacing to optimize the selectivity-permeability trade-off near the Gaussian peak at $d_{\text{spacing}} \approx 5$ Å.

A.13 Williams-Otto Process

We consider the Williams-Otto process [84, 85], a benchmark chemical process consisting of a continuously stirred tank reactor (CSTR) with three reactions:

$$\mathrm{A} + \mathrm{B} \rightarrow \mathrm{C}, \qquad \mathrm{C} + \mathrm{B} \rightarrow \mathrm{P} + \mathrm{E}, \qquad \mathrm{P} + \mathrm{C} \rightarrow \mathrm{G},$$

where P is the desired product and G is waste. We formulate this as a multi-scale grey-box problem where catalyst properties (black-box from DFT calculations) affect kinetic selectivity, while the reactor mass balances at steady state are known white-box equations. The problem can be formulated as:

$$
\begin{aligned}
\max_{T,F_{fB},\mu,\varepsilon,\sigma_c} \quad & F_{pP} && (18)\\
\text{s.t.} \quad & k_i(T) = \frac{a_i}{\rho}\exp\left(-\frac{b_i}{T}\right), \quad i \in \{1,2,3\},\\
& r_1 = y_1 \cdot k_1(T) \cdot X_A X_B, \quad r_2 = y_2 \cdot k_2(T) \cdot X_B X_C, \quad r_3 = k_3(T) \cdot X_C X_P,\\
& 0 = F_{fA} - \mu X_A - r_1 \cdot m,\\
& 0 = F_{fB} - \mu X_B - r_1 \cdot m - r_2 \cdot m,\\
& 0 = -\mu X_C + 2r_1 \cdot m - 2r_2 \cdot m - r_3 \cdot m,\\
& 0 = -\mu X_E + 2r_2 \cdot m,\\
& 0 = -\mu X_P + r_2 \cdot m - 0.5 r_3 \cdot m,\\
& 0 = -\mu X_G + 1.5 r_3 \cdot m,\\
& F_{pP} = \mu X_P - 0.1\mu X_E, \quad F_{wG} = \mu X_G,\\
& g_1(x,y) := F_{wG} - 1.0 \leq 0,\\
& g_2(x,y) := 0.5 - X_A - X_B - X_C - X_E - X_P - X_G \leq 0,\\
& y_1 = f_1^{\mathrm{BB}}(\varepsilon,\sigma_c) := \exp\left(-\frac{(\varepsilon-15)^2}{50}\right)\cdot\left(1 + 0.1(\sigma_c - 5)\right),\\
& y_2 = f_2^{\mathrm{BB}}(\varepsilon,\sigma_c) := 1.5\cdot\exp\left(-\frac{(\varepsilon-12)^2}{40}\right)\cdot\exp\left(-\frac{(\sigma_c-6)^2}{8}\right),\\
& T \in [400, 700]\ {}^\circ\mathrm{R}, \quad F_{fB} \in [5, 50]\ \mathrm{klb/h}, \quad \mu \in [50, 200]\ \mathrm{klb/h},\\
& \varepsilon \in [5, 25]\ \mathrm{kJ/mol}, \quad \sigma_c \in [3, 10]\ \text{Å},\\
& x = [x^{\mathrm{WB}}, x^{\mathrm{BB}}], \quad x^{\mathrm{WB}} = [T, F_{fB}, \mu], \quad x^{\mathrm{BB}} = [\varepsilon, \sigma_c],
\end{aligned}
$$

where $X_i = m_i/m$ are mass fractions and m is total reactor mass. Refer to the parameters in Table 11.

The black-box outputs y_1 and y_2 represent catalyst activity factors for reactions 1 and 2, modeled as volcano-type functions of catalyst surface energy ε and pore size σ_c. The constraint g_1 limits waste production ($F_{wG} \leq 1.0$ klb/h) and g_2 ensures mass balance closure ($\sum_i X_i \geq 0.5$). The global maximum net product flow is 5.470 klb/h with $x^{\mathrm{WB}\star} = [581.9, 50.0, 50.0]^\top$, $x^{\mathrm{BB}\star} = [12.6, 6.11]^\top$.

Appendix B Acquisition Function Details

All BO experiments use Expected Improvement (EI) as the acquisition function. We also implemented four alternatives for completeness; the EI results are reported in the main text because EI is the most widely used baseline and enables direct comparison with prior work.

Table 11: Variables, physical meaning, bounds, and units (Williams-Otto Process)

Symbol	**Description**	**Range / Value**	**Units**
White-box decision variables			
T	Reactor temperature	[400, 700]	°R
F_{fB}	Feed flow rate of B	[5, 50]	klb/h
μ	Total outlet flow rate	[50, 200]	klb/h
Black-box decision variables			
ε	Catalyst surface energy	[5, 25]	kJ/mol
σ_c	Catalyst pore size	[3, 10]	Å
Black-box outputs			
y_1	Activity factor for reaction 1 ($A + B \to C$)	—	—
y_2	Activity factor for reaction 2 ($C + B \to P + E$)	—	—
Fixed parameters			
a_1	Pre-exponential factor (reaction 1)	5.9755×10^9	h^{-1}
a_2	Pre-exponential factor (reaction 2)	2.5962×10^{12}	h^{-1}
a_3	Pre-exponential factor (reaction 3)	9.6283×10^{15}	h^{-1}
b_1	Activation temperature (reaction 1)	12000	°R
b_2	Activation temperature (reaction 2)	15000	°R
b_3	Activation temperature (reaction 3)	20000	°R
ρ	Density	50	lb/ft^3
F_{fA}	Feed flow rate of A (fixed)	10	klb/h

B.1 Expected Improvement

For minimization, Expected Improvement is defined as

$$\alpha_{\mathrm{EI}}(x) = (\hat{y} - \mu(x) - \xi)\,\Phi(Z) + \sigma(x)\,\phi(Z), \quad Z = \frac{\hat{y} - \mu(x) - \xi}{\sigma(x)}, \tag{19}$$

where $\mu(x)$ and $\sigma(x)$ are the GP posterior mean and standard deviation, $\hat{y}$ is the best observed objective value, Φ and ϕ are the standard normal CDF and PDF, and $\xi \geq 0$ is an exploration parameter. When $\sigma(x) = 0$, we set $\alpha_{\mathrm{EI}}(x) = 0$. For maximization, the improvement direction is reversed: $Z = (\mu(x) - \hat{y} - \xi)/\sigma(x)$.

The exploration parameter ξ controls the trade-off between exploitation ($\xi = 0$, selecting points where the GP mean is low) and exploration ($\xi > 0$, favoring regions of high uncertainty even if the mean is not promising). We sweep $\xi \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$ in all experiments.

B.2 Probability of Improvement

Probability of Improvement measures the probability that a candidate point improves over the current best:

$$\alpha_{\mathrm{PI}}(x) = \Phi(Z), \quad Z = \frac{\hat{y} - \mu(x) - \xi}{\sigma(x)}. \tag{20}$$

PI is a simpler alternative to EI that ignores the magnitude of improvement. It tends to exploit more aggressively than EI for the same ξ.

B.3 Lower Confidence Bound

The Lower Confidence Bound (LCB) acquisition function for minimization is

$$\alpha_{\mathrm{LCB}}(x) = -\big(\mu(x) - \kappa\,\sigma(x)\big), \tag{21}$$

where $\kappa > 0$ controls exploration. We negate LCB so that all acquisition functions follow the convention that higher values are better (i.e., the next point is selected by $\arg\max_x \alpha(x)$). For maximization, the Upper Confidence Bound $\alpha_{\mathrm{UCB}}(x) = \mu(x) + \kappa\,\sigma(x)$ is used. We set $\kappa = 2.0$ by default.

B.4 Modified Watson-Barnes 2

The modified Watson-Barnes 2 (mWB2) acquisition function was proposed in the COBALT framework [6]:

$$\alpha_{\mathrm{mWB2}}(x) = s_n \cdot \alpha_{\mathrm{EI}}(x) - \mu(x), \quad s_n = -3\left(1 - \frac{n}{N}\right), \tag{22}$$

where n is the current iteration and N is the total budget. The dynamic scaling factor s_n starts large (emphasizing EI-driven exploration) and decreases toward zero (shifting weight to the GP mean for exploitation). For maximization, the sign of $\mu(x)$ is reversed.

B.5 Thompson Sampling

Thompson Sampling draws a sample function $\tilde{f}$ from the GP posterior and selects the point that optimizes the sample:

$$x_{\text{next}} = \arg\min_{x \in \mathcal{X}} \tilde{f}(x), \quad \tilde{f} \sim \mathcal{GP}(\mu, k). \tag{23}$$

In practice, we evaluate $\tilde{f}$ at a discrete set of candidate points by drawing from the multivariate normal $\mathcal{N}(\mu(X_{\text{grid}}), K(X_{\text{grid}}, X_{\text{grid}}))$. Exploration arises naturally from the randomness of the posterior draw rather than from an explicit exploration parameter.

B.6 Acquisition Optimization

For all acquisition functions, we maximize $\alpha(x)$ over a random grid of 1,000 candidate points drawn uniformly from the search domain (either $\mathcal{X}^{\text{BB}}$ for bilevel BO or $\mathcal{X}^{\text{WB}} \times \mathcal{X}^{\text{BB}}$ for black-box BO). This grid search approach avoids the gradient-based optimization of the acquisition function that can be problematic in low dimensions with multi-modal acquisition landscapes. The candidate with the highest acquisition value is selected as the next evaluation point.

Appendix C Gaussian Process Configuration

Both BO solvers use `scikit-learn`'s `GaussianProcessRegressor` with identical configuration. The kernel is a product of a constant kernel and an ARD RBF kernel, plus additive white noise:

$$k(x, x') = \sigma_f^2 \prod_{d=1}^{D} \exp\left(-\frac{(x_d - x'_d)^2}{2\ell_d^2}\right) + \sigma_n^2\, \delta(x, x'), \tag{24}$$

where D is the input dimension (n_{BB} for bilevel BO, $n_{\text{WB}} + n_{\text{BB}}$ for black-box BO), σ_f^2 is the signal variance, ℓ_d is the length scale for dimension d (automatic relevance determination), and σ_n^2 is the noise variance.

Table 12 summarizes the hyperparameter bounds and optimization settings.

Table 12: GP hyperparameter configuration.

Parameter	**Value / Bounds**	**Notes**
Signal variance σ_f^2	$[10^{-10},\ 10^{3}]$	Initial value 1.0
Length scales ℓ_d	$[10^{-10},\ 10]$	One per input dimension (ARD)
Noise variance σ_n^2	$[10^{-10},\ 10^{-1}]$	Initial value 10^{-5}
Regularization α	10^{-6}	Added to diagonal for numerical stability
Normalize y	True	Observations standardized before fitting
Optimizer restarts	5	L-BFGS-B on log marginal likelihood

The bilevel BO surrogate fits a GP over $n_{\text{BB}} = 1$–2 input dimensions, while the black-box BO surrogate fits over $n_{\text{WB}} + n_{\text{BB}} = 2$–$5$ dimensions. Since GP fitting scales as $O(N^3)$ in the number of observations N (which is the same for both methods) and the per-prediction cost is $O(N^2D)$, the lower dimensionality of bilevel BO reduces prediction cost but does not change the dominant $O(N^3)$ training cost. In practice, with $N \leq 250$ and $D \leq 5$, GP fitting is fast for both methods (typically <0.1 seconds per iteration).

Appendix D DE Baseline Details

The black-box DE baseline solves the full problem (4) directly using SciPy's `differential_evolution` over the full variable space $[x^{\mathrm{WB}}, x^{\mathrm{BB}}]$ with a population size multiplier of 15, a maximum of 1,000 generations, and L-BFGS-B polishing of the best solution.

D.1 Constraint Handling

Constraints are handled via SciPy's `NonlinearConstraint` interface, enforcing $g(x^{\mathrm{WB}}, y, x^{\mathrm{BB}}) \leq 0$. After optimization, feasibility is verified: solutions satisfying all constraints to within 10^{-6} are considered feasible. If no feasible solution is found, the solver returns the solution with minimum constraint violation.

In contrast, both BO methods track feasibility explicitly. For black-box BO, constraints are handled via a penalty of 10^6 added to the objective when any $g_i(x) > 0$. For bilevel BO, the inner DE solver checks constraint satisfaction and returns a penalty value when the inner problem is infeasible for a given y.

D.2 Sample Efficiency Comparison

Table 13 compares the number of black-box evaluations across solvers. The DE baseline evaluates f^{BB} at every candidate in the population at every generation, resulting in substantially more f^{BB} evaluations than either BO method.

Table 13: Black-box evaluations (f^{BB} calls) per solver.

Solver	**f^{BB} evaluations**	**Notes**
Black-box DE	population-dependent	$15 \times n_{\mathrm{dim}}$ per generation
Black-box BO	$n_{\mathrm{init}} + 200$	One f^{BB} per iteration
Bilevel BO	$n_{\mathrm{init}} + 200$	One f^{BB} per iteration

Both BO methods call f^{BB} exactly once per iteration, so the total sample budget is $n_{\mathrm{init}} + N$ where $N = 200$ is the number of BO iterations. With $n_{\mathrm{init}} = 5$ (the default), this gives 205 evaluations. Bilevel BO additionally solves an inner DE at each iteration, but this uses only the white-box model f^{WB} (which is cheap) and does not require additional calls to f^{BB}.

D.3 Failure Modes

The DE baseline fails on two problems. On Rastrigin, multimodality in the full combined space challenges even global solvers: the DE baseline achieves a final regret of 3.38, worse than both BO methods. On Small-Feasible-Region-2, the narrow feasible region limits convergence: the DE baseline reports a regret of 0.055, while bilevel BO achieves effectively zero regret.

Appendix E Global Optimum Verification

Each problem's global optimum $J^\star$ was verified through a multi-stage process designed to provide high confidence that the reported optima are correct.

E.1 Verification Procedure

We used two complementary approaches, each repeated across 10 independent random seeds. First, we ran differential evolution (DE) over the full variable space $[x^{\mathrm{WB}}, x^{\mathrm{BB}}]$ with a population size of 30, a maximum of 2,000 generations, tight convergence tolerances (tol = atol = 10^{-12}), and SLSQP polishing of the best solution. Constraints were handled via `NonlinearConstraint`. Second, we ran multi-start SLSQP with 500 Latin hypercube starting points per seed.

For each problem, we compared the best feasible solutions across all 20 runs (10 DE + 10 multi-start NLP). All 13 problems achieved agreement to within 2×10^{-4} relative error, confirming the reported optima. For constrained problems, we verified that the optimal solution satisfies all inequality constraints $g_i(x^{\mathrm{WB}\star}, y^\star) \leq 0$ to within a tolerance of 10^{-6}.

E.2 Verified Optima

Table 14 reports the verified global optimum for each problem, along with the optimal decision variables.

Table 14: Verified global optima for all 13 benchmark problems. Optima verified via differential evolution (popsize = 30, maxiter = 2000, 10 seeds) and multi-start SLSQP (500 Latin hypercube starts, 10 seeds).

Problem	$J^\star$	$x^{\mathrm{WB}\star}$	$x^{\mathrm{BB}\star}$
Small-Feasible-Region 1	0.2532	[4.712]	[1.253]
Small-Feasible-Region 2	−2.000	[4.712]	[5.500]
Rastrigin	0.0000	[0, 0]	[0]
Toy-Hydrology	0.5998	[0.405]	[0.195]
Rosen-Suzuki	−44.00	[0, 1]	[2, −1]
CSTR	92.32%	[647.3, 20.0, 5.0]	[−1.047, 0.200]
Heat-Exchanger	1000.0	[1.0, 0.1, 19.3]	[50.0, 0.0]
PSA	−0.6433	[3.97, 0.1, 10.0]	[6.04, 19.6]
Batch-Reactor	−1.749	[500, 4.39, 5.0]	[−1.01, 2.00]
Distillation	11,758	[1.0, 11.0, 0.5]	[19.8, 9.99]
Evaporator	−1.959	[4.09, 19.1, 5.0]	[120.5, 2.0]
Membrane	11.00	[10.0, 0.1, 1.0]	[0.30, 5.13]
Williams-Otto	5.470	[581.9, 50.0, 50.0]	[12.6, 6.11]

Appendix F Proof of Proposition 1

Proof. We show that inf(bilevel) = inf(original) by constructing mappings in both directions.

(i) Any feasible point of (4) *maps to a feasible outer-level point with equal or worse objective.* Let $(x^{\mathrm{WB}}, x^{\mathrm{BB}})$ be feasible for (4) with $y = f^{\mathrm{BB}}(x^{\mathrm{BB}})$. By Assumption 1(ii), the inner problem (5c)–(5f) at this x^{BB} and y has a global solution $x^{\mathrm{WB}\star}$ satisfying $J(x^{\mathrm{WB}\star}, y, x^{\mathrm{BB}}) \leq J(x^{\mathrm{WB}}, y, x^{\mathrm{BB}})$. Thus x^{BB} is outer-feasible with objective value no worse than the original point.

(ii) Any outer-level feasible point maps back to a feasible point of (4) *with equal objective.* Let $\tilde{x}^{\mathrm{BB}}$ be outer-feasible with $\tilde{y} = f^{\mathrm{BB}}(\tilde{x}^{\mathrm{BB}})$ and inner solution $\tilde{x}^{\mathrm{WB}\star}$. By construction, $\tilde{x}^{\mathrm{WB}\star}$ satisfies $f^{\mathrm{WB}}(\tilde{x}^{\mathrm{WB}\star}, \tilde{y}) = 0$, $g(\tilde{x}^{\mathrm{WB}\star}, \tilde{y}, \tilde{x}^{\mathrm{BB}}) \leq 0$, and $\tilde{x}^{\mathrm{WB}\star} \in \mathcal{X}^{\mathrm{WB}}$. Together with $\tilde{x}^{\mathrm{BB}} \in \mathcal{X}^{\mathrm{BB}}$, the pair $(\tilde{x}^{\mathrm{WB}\star}, \tilde{x}^{\mathrm{BB}})$ is feasible for (4) with objective $J(\tilde{x}^{\mathrm{WB}\star}, \tilde{y}, \tilde{x}^{\mathrm{BB}})$.

(iii) Infeasibility. If the inner problem is infeasible for some $\tilde{y} \in \mathrm{range}(f^{\mathrm{BB}})$—i.e., no $x^{\mathrm{WB}} \in \mathcal{X}^{\mathrm{WB}}$ satisfies $f^{\mathrm{WB}}(x^{\mathrm{WB}}, \tilde{y}) = 0$ and $g(x^{\mathrm{WB}}, \tilde{y}, \tilde{x}^{\mathrm{BB}}) \leq 0$—then no feasible point of (4) exists at this $\tilde{x}^{\mathrm{BB}}$ either, so excluding it from the outer search does not lose optimality.

From (i) and (ii), the feasible sets project onto each other with preserved objectives, so inf(bilevel) = inf(original). To see how a global optimum of (4) appears in (5): let $(x^{\mathrm{WB}\star\star}, x^{\mathrm{BB}\star\star})$ be a global minimizer of (4) with $y^{\star\star} = f^{\mathrm{BB}}(x^{\mathrm{BB}\star\star})$. By step (i), $x^{\mathrm{BB}\star\star}$ is outer-feasible in (5), and the inner problem at $x^{\mathrm{BB}\star\star}$ returns some $\hat{x}^{\mathrm{WB}}$ with $J(\hat{x}^{\mathrm{WB}}, y^{\star\star}, x^{\mathrm{BB}\star\star}) \leq J(x^{\mathrm{WB}\star\star}, y^{\star\star}, x^{\mathrm{BB}\star\star})$. Since $(x^{\mathrm{WB}\star\star}, x^{\mathrm{BB}\star\star})$ is already globally optimal for (4), the inequality must hold with equality: $\hat{x}^{\mathrm{WB}} = x^{\mathrm{WB}\star\star}$ (up to alternative optima with the same objective value). Thus the bilevel formulation recovers both the optimal $x^{\mathrm{BB}\star\star}$ at the outer level and the optimal $x^{\mathrm{WB}\star\star}$ at the inner level.

Assumption 1(ii) is used in step (i) to ensure the inner global optimum is attained, and in step (ii) to ensure the inner solution is globally optimal (not merely a local minimum). The surrogate in Algorithm 1 models $J(x^{\mathrm{WB}\star}(x^{\mathrm{BB}}), f^{\mathrm{BB}}(x^{\mathrm{BB}}), x^{\mathrm{BB}}) : \mathbb{R}^{n_{\mathrm{BB}}} \times \mathbb{R}^{n_y} \times \mathbb{R}^{n_{\mathrm{WB}}} \mapsto \mathbb{R}$, which is a function of n_{BB} variables. □

Appendix G Implementation Details

Infeasibility handling. If the inner problem is infeasible for some y—because the black-box outputs lead to a white-box problem with no feasible solution—we return a large penalty value. The acquisition function naturally steers away from these regions without requiring explicit constraint modeling at the outer level.

Modularity. The inner optimizer can be swapped (DE, SLSQP, IPOPT, BARON) without changing the outer BO framework. Similarly, any acquisition function (EI, PI, LCB) can be used in the outer loop.

Table 15: Iterations to reach 1% of initial regret. ">200" means the target was not reached within 200 iterations. Speedup values marked with $\geq$ are lower bounds (BB-BO did not converge within 200 iterations).

Problem	**BB-BO**	**Bi-BO**	**Speedup**
Small-Feasible-Region	>200	>200	—
Small-Feasible-Region-2	>200	36	$\geq$5.6×
Rastrigin	>200	50	$\geq$4.1×
Toy-Hydrology	>200	14	$\geq$14.9×
Rosen-Suzuki	>200	>200	—
CSTR	>200	14	$\geq$13.9×
Heat-Exchanger	151	102	1.5×
PSA	>200	>200	—
Batch-Reactor	>200	>200	—
Distillation	>200	10	$\geq$20.1×
Evaporator	>200	>200	—
Membrane	>200	>200	—
Williams-Otto	>200	9	$\geq$22.3×

Appendix H Convergence Speed Details

Appendix I Hyperparameter Sensitivity Details

We sweep over two key BO hyperparameters: the number of initial samples $n_{\text{init}} \in \{1, 5, 20, 50\}$ drawn from a Latin hypercube design, and the EI exploration parameter $\xi \in \{0.001, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}$. Each configuration is repeated 10 times with independent seeds 42–51 to quantify variability. Every BO run uses 200 iterations after initialization. In total, the experiment comprises 7,410 independent optimization runs: $13 \times 4 \times 7 \times 10 \times 2 = 7{,}280$ BO runs (black-box and bilevel) plus $13 \times 10 = 130$ NLP runs (one per problem-seed combination, since NLP results do not depend on n_{init} or ξ).

Table 16: Bilevel BO final regret across n_{init} values (best ξ per problem).

Problem	$n_{\text{init}} = 1$	$n_{\text{init}} = 5$	$n_{\text{init}} = 20$	$n_{\text{init}} = 50$
Small-Feasible-Region	0.0345	0.0420	0.0449	0.0286
Small-Feasible-Region-2	0.0000	0.0000	0.0000	0.0000
Rastrigin	0.0001	0.0001	0.0001	0.0000
Toy-Hydrology	0.0000	0.0000	0.0000	0.0000
Rosen-Suzuki	0.3909	0.4763	0.5257	0.3520
CSTR	0.0052	0.0052	0.0053	0.0050
Heat-Exchanger	0.0002	0.0002	0.0002	0.0002
PSA	0.0115	0.0046	0.0058	0.0061
Batch-Reactor	0.0282	0.0354	0.0314	0.0372
Distillation	0.0328	0.0321	0.0321	0.0321
Evaporator	0.0345	0.0393	0.0472	0.0781
Membrane	0.0001	0.0001	0.0001	0.0001
Williams-Otto	0.0000	0.0000	0.0000	0.0000

Table 17: Best EI exploration parameter ξ per problem, selected by lowest median final regret for bilevel BO across all n_{init} values. These are the ξ values used to report results in Table 3.

Problem	**best ξ**
Small-Feasible-Region	0.5
Small-Feasible-Region-2	0.01
Rastrigin	0.2
Toy-Hydrology	0.001
Rosen-Suzuki	0.2
CSTR	0.001
Heat-Exchanger	0.01
PSA	0.2
Batch-Reactor	0.01
Distillation	0.001
Evaporator	0.1
Membrane	0.5
Williams-Otto	0.001

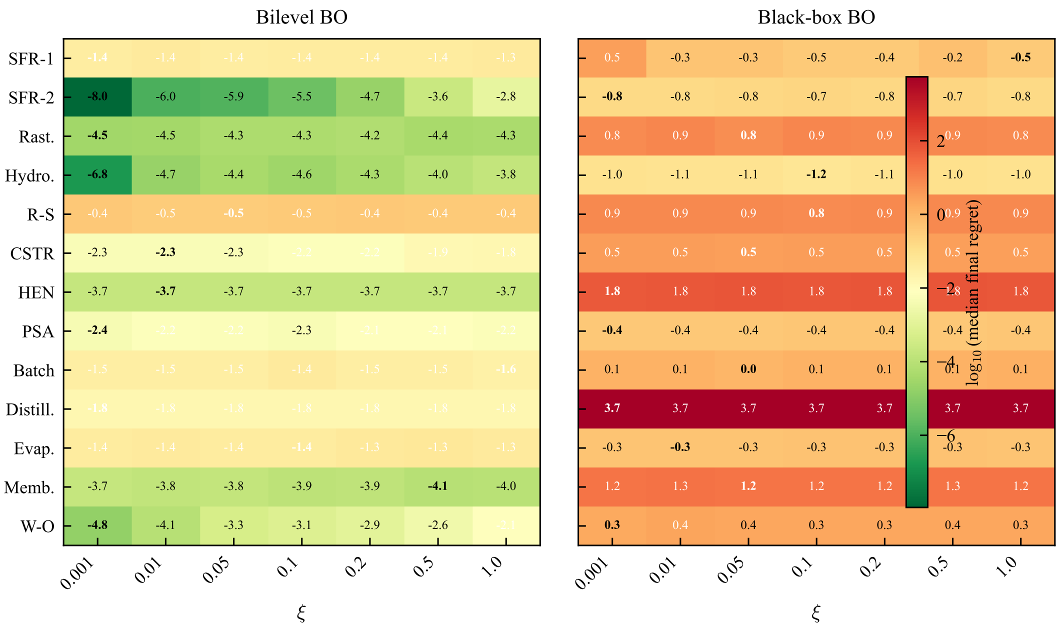

Figure 11: Sensitivity to the EI exploration parameter ξ. Left: bilevel BO; right: black-box BO. Rows are problems, columns are ξ values, aggregated over all n_{init}. Cell values show $\log_{10}$(median final regret); bold indicates best ξ per problem. No single ξ is universally optimal, but bilevel BO outperforms BB-BO regardless of ξ.

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